

Universal Control Network Installation

UN20-500

Installation/Universal Control Network

Universal Control Network Installation

UN20-500

R684

CE Compliant

03/2013

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About This Publication

This publication provides the necessary information required to install and checkout a Universal Control Network (UCN).

This publication supports **TotalPlant** Solution (TPS) Network Release 684 and earlier software releases. TPS is the evolution of TDC 3000^X.

This publication supports CE Compliant equipment. Any equipment designated as “CE Compliant” complies with the European Union EMC and Health and Safety Directives. All equipment shipping into European Union countries after January 1, 1996 requires this type of compliance—denoted by the “CE Mark.”

ATTENTION

With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network.

Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms EHPM, ENIM, and EUCN respectively.

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Standard Symbols

Scope

The standard symbols used in this publication are defined as follows.

ATTENTION

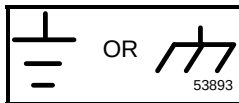
Notes inform the reader about information that is required, but not immediately evident.

CAUTION

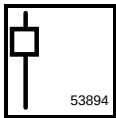
Cautions tell the user that damage may occur to equipment if proper care is not exercised.

WARNING

Warnings tell the reader that potential personal harm or serious economic loss may happen if instructions are not followed.



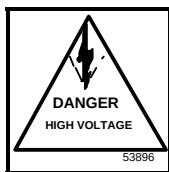
Ground connection to building safety ground.



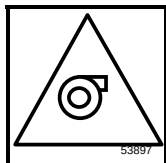
Ground stake for building safety ground.



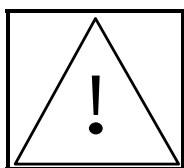
Electrical Shock Hazard—can be lethal.



Electrical Shock Hazard—can be lethal.



Rotating Fan—can cause personal injury.



Caution—refer to the appropriate installation document.

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Acronyms

APM	Advanced Process Manager
HPM	High-Performance Process Manager
EHPM.....	Enhanced High-Performance Process Manager
ENIM	Enhanced Network Interface Module
FTE	Fault Tolerant Ethernet
EUCN.....	Enhanced Universal Control Network
LM	Logic Manager
MRG.....	Master Reference Ground
NIM	Network Interface Module
PM.....	Process Manager
SM.....	Safety Manager
UCN	Universal Control Network
UCNE.....	Universal Control Network Extender

References

Publication Title	Publication Number	Binder Title	Binder Number
<i>TPS System Site Planning</i>	SW02-550	System Site Planning - 1	TPS 3020-1
<i>Universal Control Network (UCN) Planning</i>	UN02-501	System Site Planning - 1	TPS 3020-1
<i>Universal Control Network Guidelines</i>	UN12-510	Installation/Universal Control Network	TPS 3041
<i>Process Manager/Advanced Process Manager Planning</i>	PM02-501	System Site Planning - 1	TPS 3020-1
<i>Process Manager/Advanced Process Manager Installation</i>	PM20-501	Implementation/PM/APM	TPS 3043
<i>Process Manager/Advanced Process Manager Service</i>	PM13-501	PM/APM/HPM Service - 1	TPS 3061-1
<i>High-Performance Process Manager Planning</i>	HP02-500	System Site Planning - 2	TPS 3020-2
<i>High-Performance Process Manager Installation</i>	HP20-50	Implementation/HPM -3	TPS 3066-3
<i>High-Performance Process Manager Service</i>	HP13-500	PM/APM/HPM Service - 1	TPS 3061-1
<i>Logic Manager Planning</i>	LM02-501	System Site Planning - 2	TPS 3020-2
<i>Logic Manager Installation</i>	LM20-500	Implementation/Logic Manager	TPS 3070
<i>Logic Manager Service</i>	LM13-500	LM Service	TPS 3073
<i>Safety Manager Module Installation Guide</i>	SM20-500	Implementation/Safety Manager	TPS 3074
<i>Safety Manager Module Service</i>	SM13-500	Implementation/Safety Manager	TPS 3074

Section 1 – Introduction

1.1 Overview

Introduction

This manual is used for the installation of a Universal Control Network (UCN) or an Enhanced Universal Control Network (EUCN) at a site. The UCN provides a coaxial cable and optional fiber optic cable system for the interconnection of Process Manager (PM), Advanced Process Manager (APM), High-Performance Process Manager (HPM), Logic Manager (LM), Safety Manager (SM), and Network Interface Module (NIM) network devices. With R684, EUCN provides a copper FTE cable and an optional fiber optic cable system for the connections between the Enhanced High-Performance Process Manager (EHPM), and the Enhanced Network Interface Module (ENIM) network devices. This manual is supplemented with information in the *Universal Control Network Site Planning* and *Universal Control Network Guidelines* manuals that concerns power, grounding, cabling, and the equipment environment.

Purpose of manual

The primary purpose of this manual is to provide guidance in the installation and checkout of the UCN / EUCN cable system.

Installation and checkout of network devices

The installation and checkout instructions for the Network Interface Module, Enhanced Network Interface Module (ENIM), Process Manager, Advanced Process Manager, High-Performance Process Manager, Enhanced High-Performance Process Manager (EHPM), Logic Manager, and Safety Manager network devices, or the installation of the Local Control Network (LCN) operator consoles are not provided in this manual. See the related documentation sections in the appropriate manuals for information concerning these topics.

ATTENTION

ATTENTION—For brevity, this document references only the Process Manager, Advanced Process Manager, High-Performance Process Manager, and the Enhanced High-Performance Process Manager (EHPM) in most discussions, but the discussions will in general also apply for the Logic Manager and Safety Manager.

1.2 Term Definitions

Introduction	The terms used in this document are defined as follows.
Network device	A network device is any electronic equipment that can be resident on the Universal Control Network. The equipment includes the Network Interface Module, the Process Manager, the Advanced Process Manager, the High-Performance Process Manager, the Logic Manager, and the Safety Manager. The Network Interface Module is always resident on the network because it provides the interface to the Local Control Network.
UCN trunk cable	One of two coaxial cables, designated A or B, that provides the transmission media for the UCN. The trunk cable consists of segments connected together by drop cable taps.
UCN trunk segment	The portion of the UCN trunk cable separated by drop cable taps.
UCN cable A	One of a redundant pair of UCN coaxial trunk cables that are diversely routed to reduce the possibility of simultaneous damage to the cables.
UCN cable B	One of a redundant pair of UCN coaxial trunk cables that are diversely routed to reduce the possibility of simultaneous damage to the cables.
EUCN cable A	One of a redundant pair of EUCN copper FTE cables that are diversely routed to reduce the possibility of simultaneous damage to the network media.
EUCN cable B	One of a redundant pair of EUCN copper FTE cables that are diversely routed to reduce the possibility of simultaneous damage to the network media.
Drop cable tap	The component that adapts the network device to the UCN trunk cable by means of a drop cable while maintaining the impedance of the trunk cable. Three types of drop cable taps provide connections for two, four, or eight drop cables.
CF9 Switch	The eight-port, Control Firewall (CF9) serves as the Fault Tolerant Ethernet (FTE) communication center for a given EHPM network. It consists of a Control Firewall CF9 Ethernet Switch module that plugs into its associated CF9 input/output termination assembly (IOTA).

Drop cable tap port	An F-type connector on the drop cable tap that provides connection for a trunk cable or a drop cable.
Drop cable	The coaxial cable that connects the network device to the UCN trunk cable by means of the drop cable tap.
Horizontal cable	An FTE cable associated with a structured cable plant system that connects the patch panel to a wall faceplate or surface mounted box.
UCN Extender (UCNE)	A fiber optic link between coaxial trunk cable segments.
UCNE	The device that provides the fiber optic link between coaxial trunk cable segments. A pair of UCNEs is required at each end the fiber optic link.

Section 2 – Installing the UCN/EUCN

2.1 Overview

Purpose

This section discusses bulk coaxial cable testing, cable assembly, and the installation of the Universal Control Network (UCN).

With R684, this section also covers FTE cable testing, FTE cable assembly, and the installation of the Enhanced Universal Control Network (EUCN).

Introduction

Communication between the control room (Network Interface Module) and the Process Manager (PM), the Advanced Process Manager (APM), High-Performance Process Manager (HPM), Logic Manager (LM), or Safety Manager (SM) is accomplished through the Universal Control Network. See Figure 2-1.

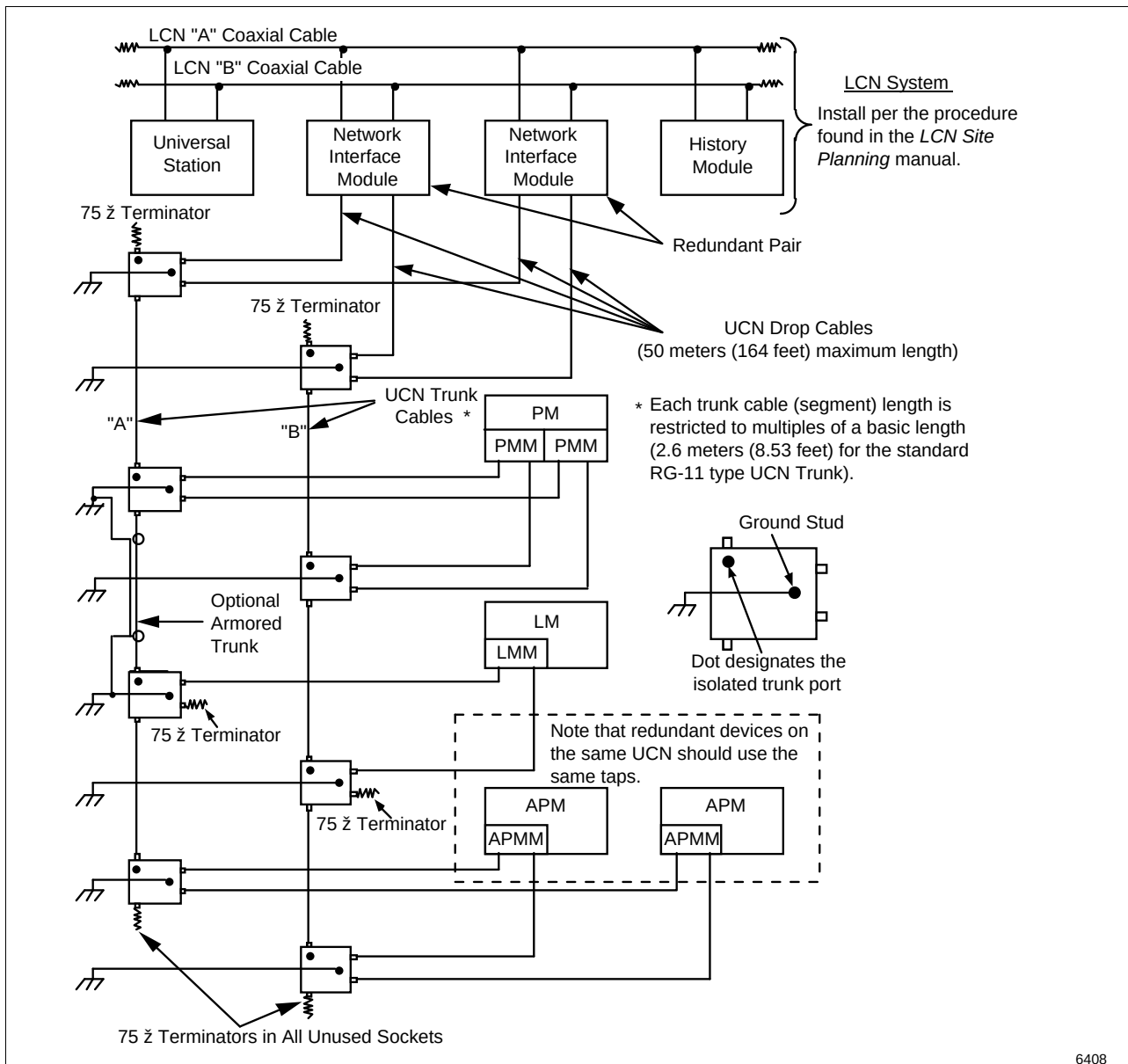
If the trunk or drop cables are to be assembled on site from reels of approved bulk cable and connector kits, complete all the instructions in subsections 2.1 through 2.3. When standard cable assemblies of fixed lengths with connectors already installed and tested are supplied by Honeywell, some portions of subsections 2.1 through 2.3 may not apply in the installation of the Universal Control Network.

With R684, EUCN systems provide communication between Enhanced Network Interface Module (ENIM) and Enhanced High-Performance Process Manager (EHPM) using Fault Tolerant Ethernet (FTE). See Figure 2-2.

2.1 Overview, Continued

UCN installation

Figure 2-1 Typical Universal Control Network Installation



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CAUTION

CAUTION—UCN coaxial cables and associated components have been designed and tested by Honeywell to meet strict standards of performance. Other coaxial cables and components have not been tested and qualified for use in the Universal Control Network. Therefore, no substitutions are approved by Honeywell. Various lengths of coaxial cable are available as standard products and must be ordered from Honeywell or Honeywell-approved sources.

The diagram illustrates a redundant LCN system architecture. At the top, the LCN System is connected to two Coaxial Cables: LCN "A" and LCN "B". These cables feed into four modules: Universal Station, Enhanced Network Interface Module (ENIM), another Enhanced Network Interface Module, and a History Module. The ENIMs are connected to a Redundant ENIM Pair. Below the LCN System, two switches, SWITCH "A" and SWITCH "B", are shown, labeled as IE-3000 Switches (Redundant Pair). These switches are connected to the LCN System and to two Control Fire Walls (CF-9 "A" and CF-9 "B"). The CF-9 units are connected to two EHPM Clusters, each consisting of two EHPM units. The connections are labeled with various FTE (Fiber Termination Equipment) levels: Copper FTE "A" Lvl 1, Copper FTE "B" Lvl 1, Copper FTE "A" Lvl 2, and Copper FTE "B" Lvl 2. A note indicates a 100 Meters Maximum on Copper Cable. Another note states: "The theoretical 'copper cable' limit for the connection from the ENIM to the EHPM is 300 meters. Practically this limit is closer to 130 meters due to the fact that the IE-3000 Switch must reside in an LCN cabinet and the Control Fire Wall (CF-9) must reside in the EHPM cabinet." A final note mentions: "Fiber FTE 'A' Lvl 2 (2 KM Maximum)" and "Fiber FTE 'B' Lvl 2 (2 KM Maximum)".

7

2.2 Bulk Coaxial Cable Test

Introduction	Test all coaxial cables before installation, especially bulk coaxial cable on a reel. If defective cables are introduced into the Universal Control Network installation, replacing a defective segment that is in service can be costly and difficult. Even though the cable has been factory tested, heat, water, crushing, bending, or dropping the reel can damage the cable before installation. Test the cable at the site to ensure its integrity.
Use a network tester	A carrier-band network tester is used to perform all tests. All Universal Control Network testing is done at 5 Mb/s (megabits/second) with a bandwidth of 1 to 30 Mhz. Return loss and attenuation are measured in dB (decibels), while signal level and noise are measured in dBmV (0 db = 1 mV). A suitable tester, model CBT, is available from Relcom Incorporated, Forest Grove, Oregon, telephone 503-357-5607.
Testable cable length	An excessively long coaxial cable length on a reel may have to be shortened to give the tester a useful end reflection. Test RG11-type trunk cable in lengths no longer than 760 meters (2500 feet). Test RG6-type drop cable in a length no longer than 300 meters (1000 feet). For other cables, consult your cable supplier for the maximum test length.
Use carrier-band terminators	Cable testing is done using precision carrier-band terminators. Do not use video or radio terminators. Carrier-band terminators are tested for 75 ohms $\pm$ 1% in the 1 to 30 Mhz carrier-band passband. The terminators supplied with the Honeywell tap kits meet these requirements.
Cable rings	Trunk cables have rings printed on the jacket. Drop cables do not. To minimize signal reflections at drop cable tap locations, the rings are spaced at 2.6 m (8.5 ft) intervals along the UCN standard RG11-type trunk cable. If a different type UCN trunk cable is used, the spacing can differ. Install connectors only at the printed ring locations on the cable's outer jacket. If your system's trunk cable does not have the printed rings on the outer jacket, it is not carrier-band cable. Do not use it. Installation instructions for the Honeywell-approved connectors on bulk coaxial cable are in Section 3.

2.2 Bulk Coaxial Cable Test, Continued

Test specifications	Install connectors on both ends of the cable on the reel and test for the following specifications.
Return loss	Install a 75-ohm terminator at one end of the cable using a coupler/splice supplied with CBT tester. Connect the other end of the cable to the tester's REC port. Avoid the use of an extension cable between the cable and the tester because it could adversely affect the test results. Test both ends for -26 dB or higher return loss at 5 Mb/s (-27 dB is better, -25 dB is bad).
Attenuation	Connect one end of the cable to the tester XMIT port and the other to the REC port. Measure the attenuation at 5 Mb/s. Again, avoid the use of extension cables between the cable and the tester. Note the cable length on the reel and calculate the dB/m attenuation. Compare the measurement to the supplier's specification and expect reasonable conformity. Do not attempt to measure small values of attenuation that are below the accuracy limits of the CBT tester.
Cable failure	A cable that fails these tests is not suitable for your system.
Cable parameters	The tester used in the tests above will not be able to certify other coax parameters, such as group delay, tilt, and impedance as required by IEEE 802.4. Use only cable that is specified by Honeywell for your carrier-band system for your cable drops and distances.
Connector removal	Remove the connectors that were installed on the cable for acceptance testing. Determine the lengths of cables needed for your system that you planned according to the <i>Universal Control Network (UCN) Planning</i> manual and cut the cables from the reel(s) to their working lengths. Then install the cables as instructed in the following subsections.

2.3 Trunk Cable Installation and Return Loss Testing

Introduction	The following installation procedures are for trunk coaxial cables. However, because drop cables can be quite long, 50 meters (164 feet), use the procedures in this subsection for drop cables when appropriate.
Use site installation drawings	Universal Control Network installation drawings for your site were generated from the layout rules in the <i>Universal Control Network (UCN) Planning</i> manual. The drawings should be similar to Figure 2-1, but with much more installation detail. Using the drawings, route the UCN trunk cables at the specified distances from high voltage and other sources of electrical interference. A mistake here may take a long time to find and correct.
UCN cable map forms	Use the UCN Cable Map forms in the <i>Universal Control Network (UCN) Planning</i> manual to record the trunk and drop cable connections to the drop cable taps.
Maximum trunk cable performance	To maximize trunk performance, fabricate all segments of trunk A from the same reel of cable. Do the same for trunk B.
Acceptable cable installation parameters	Install the cables, being careful not to mistreat the cables. Cables can be damaged during installation by exceeding the following cable installation parameters.
Maximum pull strength	Do not exceed the pull strengths listed in Table 2-1. Excessive pull permanently reduces the cable diameter, affecting impedance, and can cause electrical failure.
Maximum unsupported distance	Do not exceed the maximum unsupported distances listed in Table 2-1 without using a molded or taped messenger wire.
Minimum bend radius	Do not create a bend radius less than listed in Table 2-1 for the type of cable. Pulling the cable around a small radius permanently crushes the cable into an oval shape, affects the impedance, and can cause electrical failure.

2.3 Trunk Cable Installation and Return Loss Testing,

Continued

Cable installation parameters

Table 2-1 lists the coaxial cable installation parameters.

Table 2-1 Cable Installation Parameters

Coax Type	Maximum Pull Strength	Minimum Bend Radius	Maximum Unsupported Length
Standard RG11-Type	68 kg (150 lbs)	11 cm (4 1/2 in.)	61 m (200 ft)
Armored RG11-Type	68 kg (150 lbs)	11 cm (4 1/2 in.)	91 m (300 ft)
Standard RG6-Type	45 kg (100 lbs)	8 cm (3 in.)	61 m (200 ft)

Good installation practices

Damage to the cables can be reduced by using good practices during the installation. Limit the strain when pulling cable onto ladder racks by using pulleys over which the cable can travel. When pulling through conduit and trays, apply a lubricant. Use pull boxes at each bend and limit the distance between pull boxes to 30 meters (100 feet). If resistance is encountered while pulling the coax, stop and correct the problem before continuing.

Cable environment

Avoid attaching the cable to a structure that moves or vibrates excessively. Where there are construction cranes moving around, do not place the coax cables in the lowest overhead rack. Similarly, be careful in process areas that are prone to fire, melts, spills, floods, flares, venting, or explosions. Go around/over/underground as appropriate.

Cable installation

Once the trunk segments are in place, install the connectors at the printed ring locations on the cable's outer jacket. The assembly instructions are in Section 3.

Also, install either TRUNK A or TRUNK B markers at both ends of the cable as appropriate. They are included in the Honeywell connector kits.

After a connector has been installed, thread a protection cap from the connector kit into the connector to protect the center pin until the cable is installed.

2.3 Trunk Cable Installation and Return Loss Testing,

Continued

Return loss testing

Do not connect the drop cable taps yet. The trunk segments have to be temporarily daisy-chained and tested for any damage during their installation. Use splices/couplers available in the trunk cable splice kit and connect the trunk cable segments together to create full length trunks for both the A and B trunks. Terminate the far end of the trunk and connect the near end to the tester **REC** port. Test both ends for -26 dB or higher return loss at 5 Mb/s (-27 dB is better, -25 dB is bad). Because this is the last opportunity to test the trunk without the effects of drop cable taps and drop cables, extend the test to include the requirements of the cable assemblies tests described in subsection 2.4.

Troubleshooting

If all tests are satisfactory, proceed with the connection of the drop cable taps as described in subsection 2.4. Otherwise, investigate the problem by disconnecting the defective trunk at a splice nearest the middle of the trunk. Then test each of the two sections to locate the defective segment. By successively dividing the sections of cable in half each time, you can quickly identify the defective segment.

2.4 Cable Assembly Testing

Cable segment test procedure

Use the procedure in Table 2-2 to test any length of cable that has connectors installed at both ends.

Table 2-2 UCN Trunk Cable Segment Test Procedure

Step	Action
1	Install a carrier band 75-ohm terminator at one end of the cable assembly using a splice/coupler.
2	Connect the other end to the tester REC port and test for a return loss of -26 dB minimum (-27 dB is better, -25 dB is bad).
3	While monitoring the tester, grasp the cable, RG6-type or RG11-type, and pull on the cable with no more than 9 kg (20 lbs) stress, then release. An intermittent or constant return loss reading during this pull/release cycle implies that the connector installation is defective and has to be repaired. Repeat the test at the other end of the cable.
4	Label each end of the cable with a cable identification, such as TRUNK A. Note that colored labels, orange for cable A and brown for cable B, are to be used to help prevent cross-connecting the A and B trunk cables when installing the network trunk segments and drop cable taps. Do not release the cable without threading a protection cap on the cables connectors to protect the connector's center pin. The labels and caps are included in the Honeywell connector kit.

2.5 Drop Cable Tap Connections

Introduction

At this point, the trunk cable segments have connectors installed and have been thoroughly tested.

Drop cable installation and test procedure

Install and test the drop cable taps, and then use the procedure in Table 2-3 to install and test the drop cables.

Table 2-3 Drop Cable Installation and Test Procedure

Step	Action
1	Mount the drop cable taps (do not use the ground stud on the tap to mount the tap) and connect each tap's ground stud with a 2.5 mm ² (14 AWG) conductor to the nearest AC Safety Ground. Use any nearby grounded building frame or power distribution panel AC Safety Ground. Do not ground any cable armor at this time. See Figure 2-1.
2	Connect the trunk cables to the trunk ports at the sides of the drop cable taps. Be sure that the A trunk cables are connected only to the A taps, and the B trunk cables are connected only to the B taps. Strain relieve the trunk and drop cables near the drop cable tap ports. All drop cable taps have a white dot located next to one of the trunk ports. The port is also labeled with the term "ISOL TRUNK" on new production taps. This is the port that isolates the cable shield from ground. See Figure 2-2. When connecting trunk cables to the drop cable taps, each cable must be provided a ground at one end and isolated from ground at the other end. This is accomplished by connection of the cable to a trunk port connector without a dot at one end of the cable and a trunk port connector with a dot at the other end of the cable.

2.5 Drop Cable Tap Connections, Continued

Drop cable installation and test procedure, continued

Table 2-3 Drop Cable Installation and Test Procedure, Continued

Step	Action
3	To avoid confusion, start connecting the drop cable taps at one end of the Universal Control Network. Position the tap such that the isolated port indicator, the white dot or recessed circle, is to the right. • Connect a 75-ohm terminator to the left-hand trunk port (no dot) connector of the drop cable tap. • Connect the trunk cable to right-hand trunk port (dot) connector of the tap. • Go to the next tap and connect the other end of the cable, connected in Step b, to the left-hand trunk port (no dot) of the tap. • Connect the next trunk cable to the right-hand trunk port (dot) connector of this tap. Proceed to the next tap and repeat Steps c and d until all trunk cables have been connected. • Connect a 75-ohm terminator to the right-hand trunk port (dot) connector of the last (end) tap.
4	Tighten the trunk connectors with your fingers. Then wrench-tighten the connectors with a Honeywell-recommended torque wrench that is calibrated for a nominal 25 inch/pounds. A slight twist on the cable will indicate if the connection is sufficiently tight or not. The suggested tool is found in a Honeywell kit, model number MU-NKTQ01, part number 51109612-100. See Figure 2-3. The wrench in the kit is factory set to 25 inch/pounds. It is 152 mm (6 inches) long with a 16-mm (5/8-inch) diameter handle. Included in the kit are three hex wrench heads, sizes 11-mm (7/16-inch), 13-mm (1/2-inch), and 16-mm (5/8-inch), to be used with the torque wrench. The heads incorporate a cutout, allowing the wrench to fit over a cable while it is connected. The 13-mm (1/2-inch) flare head is used for the drop coax cable connectors, the new style trunk coax connectors, and the new style drop cable tap terminators. The 11-mm (7/16-inch) head accommodates the old style terminators used on the tap, while the 16-mm (5/8-inch) head accommodates the old style trunk cable connectors. A click can be felt or heard when the calibrated 25 inch/pounds torque limit has been reached. Do not over-tighten the connector. Avoid stress on the trunk cables at this stage. Strain-relieve each end of the trunk cable. Cables should not bend until at least 5 centimeters (2 inches) from the tap connector.

2.5 Drop Cable Tap Connections, Continued

Drop cable
installation and test
procedure, continued

Table 2-3 Drop Cable Installation and Test Procedure, Continued

Step	Action
5	Do not install the drop cables at this point. Instead, install 75-ohm terminators on all the drop cable tap ports. Also, install terminators at the end trunk ports, on the last drop cable taps in the system. Make a 2.6-meter (8.5-foot) extension cable from the same reel of cable used to fabricate the trunk cable segments and use it to connect the nearest trunk port to the tester REC port. The return loss should measure -23 dB or greater at 5 Mb/s (-24 dB is better, -22 dB is bad). Notice that this figure is 3 dB less than the -26 dB figure specified earlier for the bulk coaxial cable while on the reel. The 3 dB loss is the allowance for the drop cable taps. Repeat the test from the other end of the trunk cable.
6	If the system fails to meet the -23 dB return loss test requirement, one of the drop cable taps or a drop cable is probably defective, or a cable connection is loose. If there is a failure, disconnect the problem trunk cable at the tap nearest the middle of the trunk. Terminate portions of the break in the trunk with a splice/coupler and a 75-ohm terminator, and test both trunk halves to determine which half is defective. Successively divide the defective trunk portion until the defective tap or cable is located.
7	The trunk and all the drop cable taps have now been tested. Ground the armor of any armor cable, using clips as shown in Figure 2-4. The ground clips install at each end of the armor according to the manufacturer's instructions. Do not ground the armor to the adjacent taps. Instead, ground the armor to the point you have chosen to ground the drop cable taps. A suitable armor clip is the Scotchlok® 4460 shield connector manufactured by the 3M Company.

2.5 Drop Cable Tap Connections, Continued

Drop cable
installation and test
procedure, continued

Table 2-3 Drop Cable Installation and Test Procedure, Continued

Step	Action
8	Complete the installation of the Universal Control Network as illustrated in Figure 2-1, being sure you accomplish the following. • Ground any cable armor at both ends as illustrated in Figure 2-1. • Connect all the drop cables to their network devices. An example is the connection of the drop cables to the Process Manager or Advanced Process Manager as illustrated in Figure 2-5. Avoid crossing cable A with cable B. Pay attention to the colored markers at each end of each cable. • Terminate all unused ports as illustrated in Figure 2-1. Unused network device node connections do not require terminators. The node connection is internally terminated. Unused drop cable tap ports normally have a 75-ohm terminator connected. However, a drop cable connected to the unused tap port with a splice/coupler and terminator installed at the end of the drop cable is acceptable. • Properly tighten all cable connectors and terminators. Finger tighten the connector and then wrench-tighten it the additional amount to 25 inch/pounds with a torque wrench described previously. Twist the cable slightly to check the connection. If loose, retighten the connector. Note: The F-type connector on the UCN drop cables used in an FSC-SM node is replaced by a new F-connector (a component of the cable assembly bearing part number: 4010242) with a revised torque setting. UCN drop cables with the new F-type connectors must be tightened to a torque of 12 -15 inch/pounds. The new F-type connectors must not be tightened to the original torque setting as this may damage the connector. Newly ordered cables have a warning that specifies the new torque setting.
9	The drop cables and taps are tested together for return loss. Disconnect each drop cable at the node end (PM, APM, LM, or NIM) of the drop cable one at a time, and connect the cable to the tester REC port. Test for a return loss of -14 dB minimum (-15 dB is better, -13 dB is bad). Reconnect the tested drop cable before proceeding to the next drop cable. Open connections interfere with this test. Should a particular drop cable fail this return loss test, test the drop cable without the tap before replacing it. Disconnect it from the tap, install a 75-ohm terminator by using a splice/coupler, and test for a minimum return loss of -26 dB (-27 dB is better, -25 dB is bad). If the test result is acceptable, replace the tap. When testing of all the drop cables is complete, recheck all the connections for tightness by twisting the individual cables slightly at each connector. If the connection is loose, tighten it.
10	Proceed with the Universal Control Network power-off noise tests as discussed in Section 5.

ATTENTION

ATTENTION—Do not place your fingers on the head of the torque wrench when gripping the wrench because you may not feel the click and overtighten the connector.

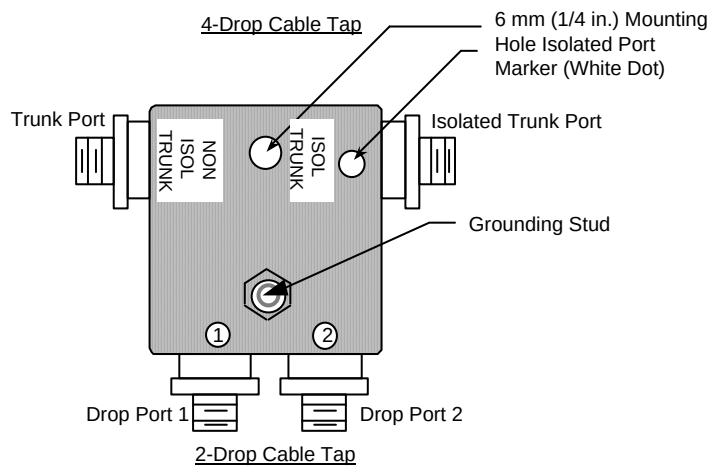
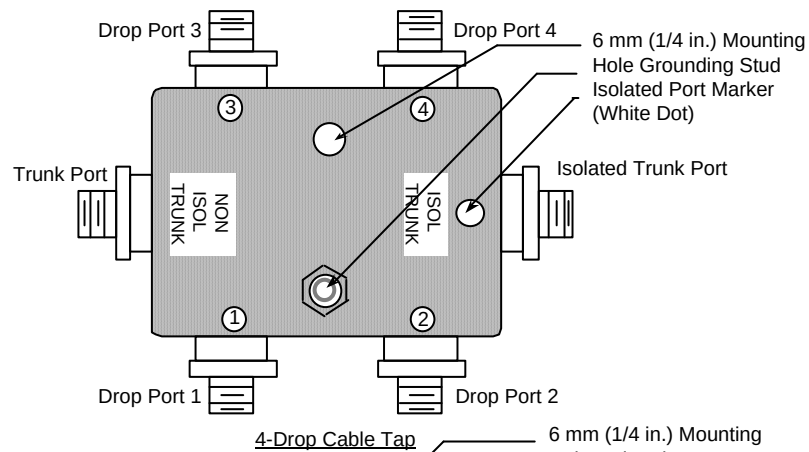
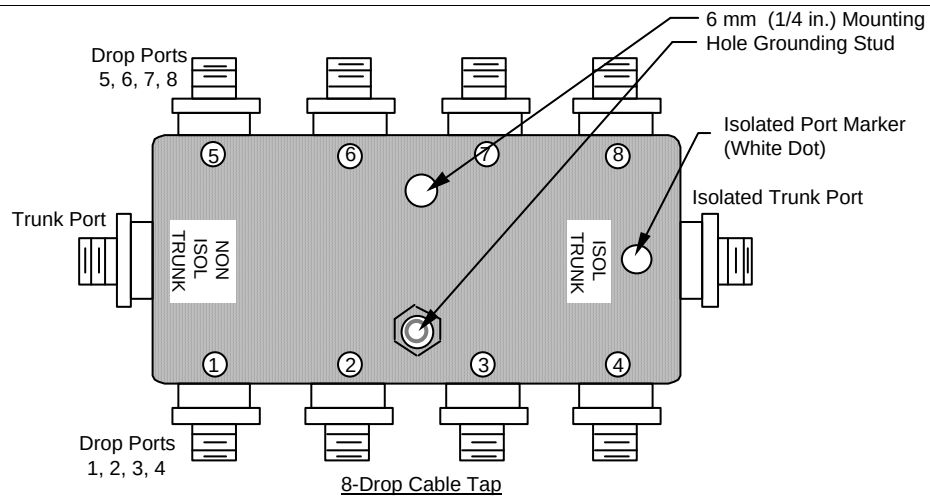
ATTENTION

ATTENTION—It is extremely important to wrench-tighten, not merely finger-tighten all F-type connectors and terminators at the drop cable taps and the network devices. Failure to do so may produce noisy connections that inhibit proper Universal Control Network performance.

2.5 Drop Cable Tap Connections, Continued

UCN drop cable taps Figure 2-2 illustrates the UCN drop cable taps that are available.

Figure 2-2 UCN Drop Cable Taps



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2.5 Drop Cable Tap Connections, Continued

UCN cable connector torque wrench

ATTENTION

Figure 2-3 illustrates the torque wrench that is used to tighten the UCN cable connectors.

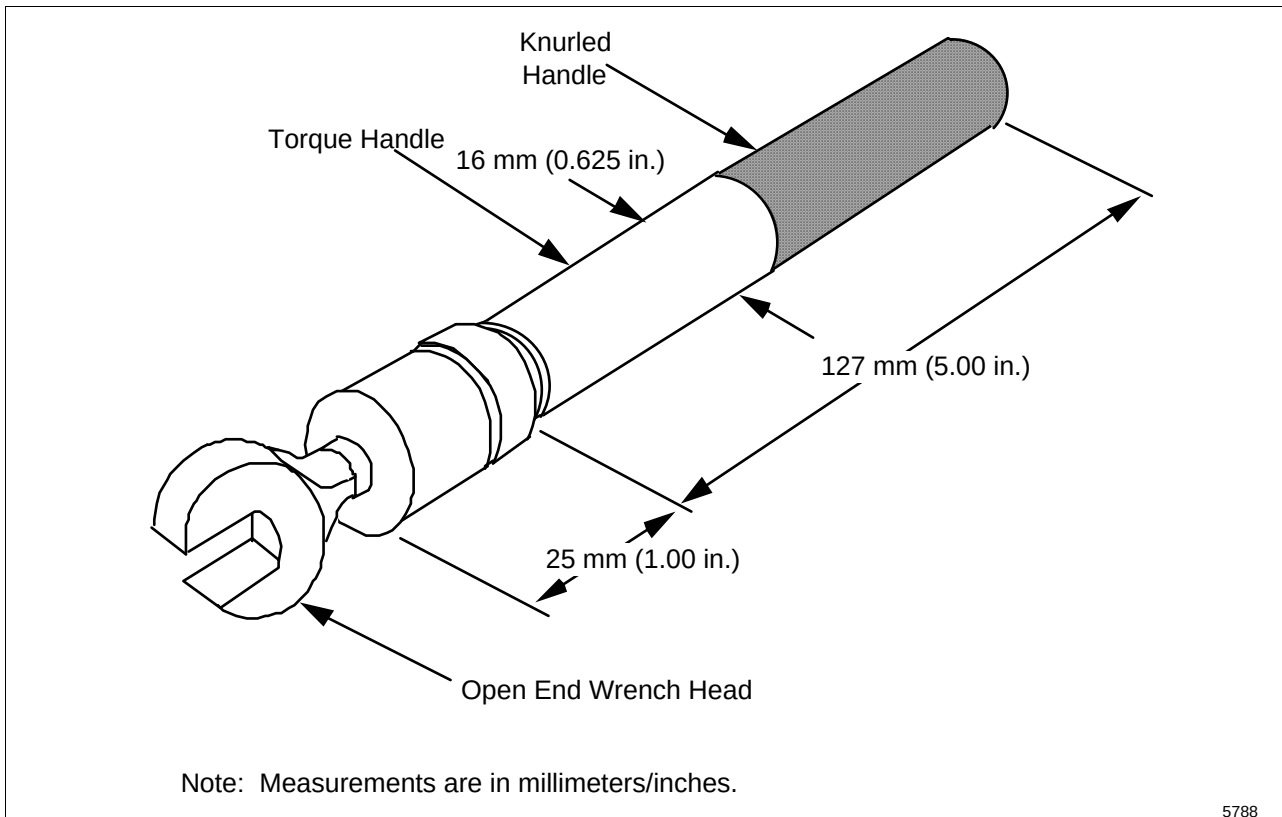
ATTENTION—The F-type connector on the UCN drop cables used in an FSC-SM node is replaced by a new F-connector (a component of the cable assembly bearing part number: 4010242) with a revised torque setting.

UCN drop cables with the new F-type connectors must be tightened to a torque of 12 -15 inch/pounds.

The new F-type connectors must not be tightened to the original torque setting as this may damage the connector.

Newly ordered cables have a warning that specifies the new torque setting.

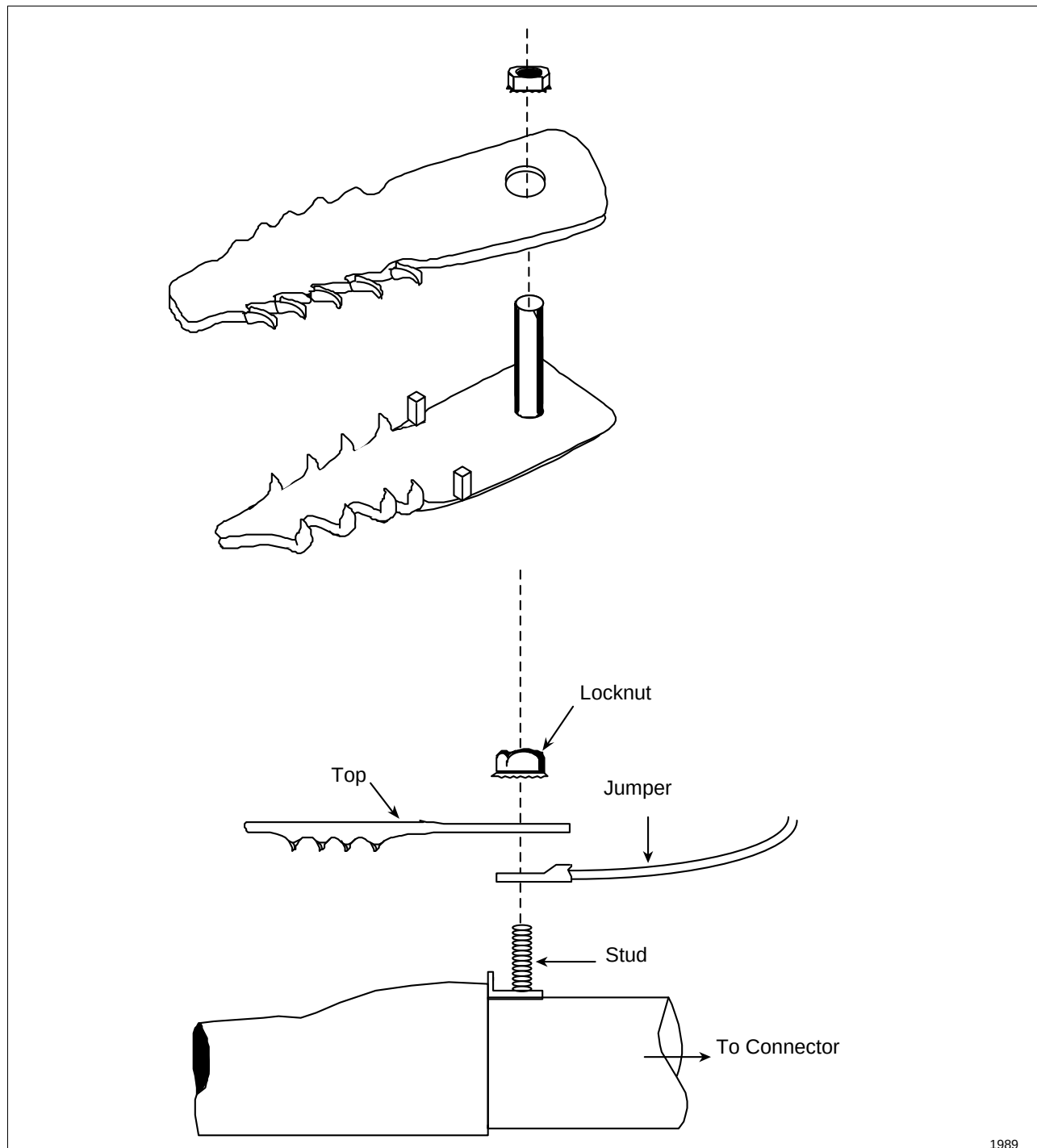
Figure 2-3 UCN Cable Connector Torque Wrench – 25 Inch/Pounds



2.5 Drop Cable Tap Connections, Continued

Armour coaxial cable clip Figure 2-4 illustrates the clip that is used with armour coaxial cable.

Figure 2-4 Armour Coaxial Cable Clip

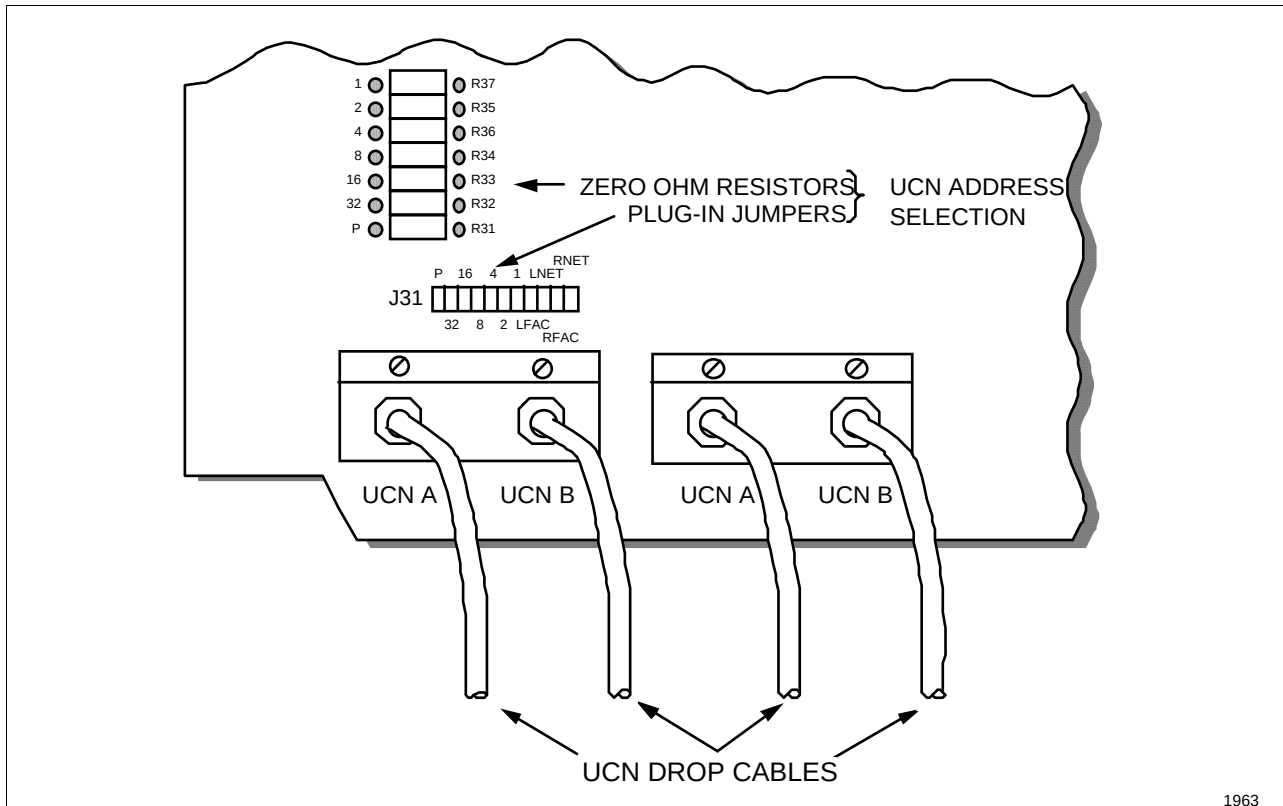


2.5 Drop Cable Tap Connections, Continued

**PMM/APMM card file
drop cable
connections**

Figure illustrates the drop cable connections to a PM or APM card file.

Figure 2-5 PM/APM Card File UCN Drop Cable Connections



2.6 EUCN Cable Construction and Installation

Cable Topology

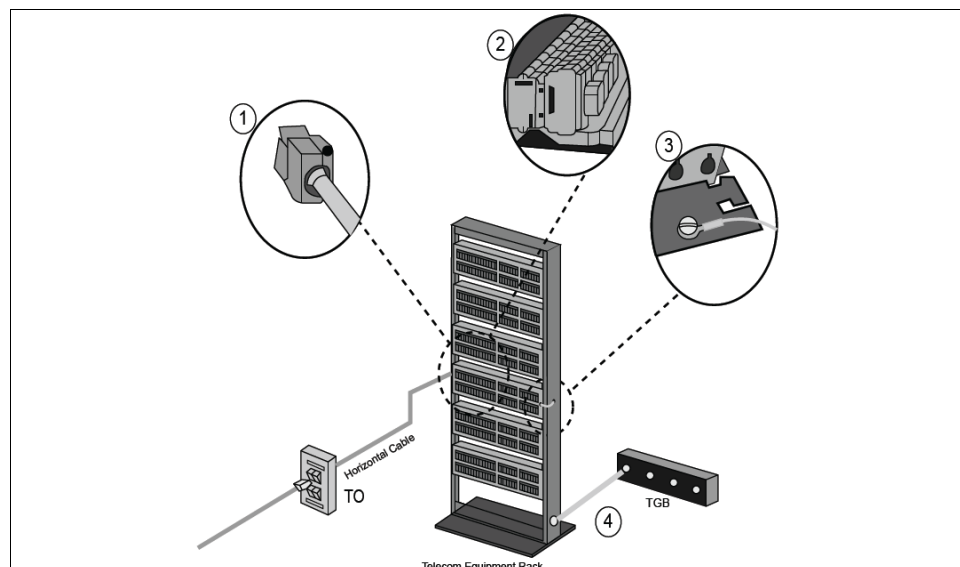
FTE cabling uses a physical star topology. Each node is cabled back to a central switch. The switch is the center of the star.

There are two network cable designs that are used for making connections between the EUCN nodes. They are:

- A structured cable plant
- An unstructured cable plant or patch cables

In a structured cable plant, there are permanently installed termination devices on either end of a horizontal cable. In the equipment room, the termination device is usually a patch panel. Patch panels come in two types. One type has the jacks permanently installed in the panel and the cable is terminated on the back of the panel, usually in a 110-style clip. The other type uses the same jack as the node outlet and the jack is snapped into a cutout in the panel. At the node end, the termination device is usually a jack installed in either a faceplate or a surface mounted box. Patch cables are used for connecting the jack to the node and the patch panel to the switch.

Figure 2-6 Telecommunications Equipment Rack



The other method is to use a long patch cable that has an RJ45 plug on each end, to directly connect the switch port to the EUCN node. It is important to note that the FTE nodes are dual connected. This means that there are two cables connecting each node to the network switches. Hence, if there are 3 nodes, then there are 6 cables that must be routed from the nodes to the network switches.

It is advisable to use a structured cable approach when:

- There are a high number of nodes in one area
- The physical cable distance from the switch to the node exceeds 65.6' (20 m)

- The nodes are separated from the switch by a physical wall

It is advisable to use patch cables when:

- The number of nodes are 6 or less
- The physical cable distance from the switch to the node is less than 33' (10 m)
- If the nodes are within the same room

When the routed cable distance between switch and node is between 10 and 20 meters, the use of a structured cable plant is always recommended by Honeywell to be the preferred option, however, if there are 6 or less nodes being connected to a switch located up to 20 meters away but in the same room, the use of patch cables is acceptable.

2.7 Selecting copper cables for the EUCN

Introduction

This section describes the characteristics of the copper patch cables and the horizontal cables that can be used in the network.

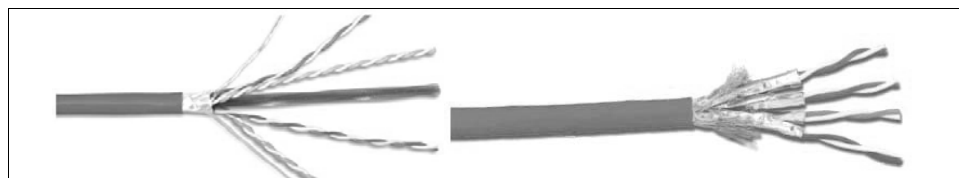
Copper data cable construction

TIA/EIA-568-B.2 specifies that the copper data cable must consist of 22 AWG to 24 AWG thermoplastic insulated solid conductors that are formed into four individually twisted-pairs enclosed by a thermoplastic jacket, and have a nominal impedance of 100 Ω . Although, the standards approve both screened and unshielded copper cable for the use of data communications, Honeywell recommends a screened cable for industrial networks. The screened type of cables, as described in the following table, has increased immunity against electromagnetic interference/radio frequency interference (EMI/RFI).

Table 2-4 Types of Screened Cables

Cable Type	Description
Screened Twisted Pair (ScTP)	The four pairs are covered with an overall fine wire screen.
Foil Twisted Pair (FTP)	The four pairs are covered with an overall metal foil shield.
Pairs in Metal Foil (PMF)	Each of the four pairs is wrapped in a metal foil shield.
Double Shield Twisted Pair (SSTP) or Shielded Twisted Pair (STP)	Each of the four pairs is wrapped in a metal foil shield which are then wrapped in an overall foil metal shield.

Figure 2-7 Screened type of Cables



Typical Unshielded Twisted
Pair cable

Shielded or Foil Twisted Pair cable

Patch cables

Patch cables that comply with the TIA/EIA-568 standard are available from Honeywell (Part number: 51305482-xxx). Honeywell supports using long patch cables for some installations if the cable meets the following requirements:

- Has a non-plenum jacket for greater flexibility
- Matches a performance category of cat 5e or better
- Has been assembled and tested at the factory. Field assembly is strongly advised against, as stated in TIA/EIA-568-B.1.10.2.4
- Has IDC RJ 45 plugs on each end
- Uses stranded conductors for increased durability and flexibility
- Is of screened construction
- Has an yellow or green jacket or connector boot
- Is 33 feet (10 meters) or less in length

Copper cable jacket types

Honeywell recommends CMR rated cable, unless the local environment requires a plenum rated cable. Typical plant environments are non-plenum or the cable is usually in conduit. A cable plant design professional can assist in determining if a plenum rated cable is required. It is recommended to use yellow and green cables for cable A and cable B, respectively.

CMR	Communications cable suitable for use in a building riser or non-plenum environment.
CMP	Communications cable suitable for use in a plenum environment where a ceiling or raised floor space is used by the air conditioning system as an air return plenum.

2.8 Terminating copper cables

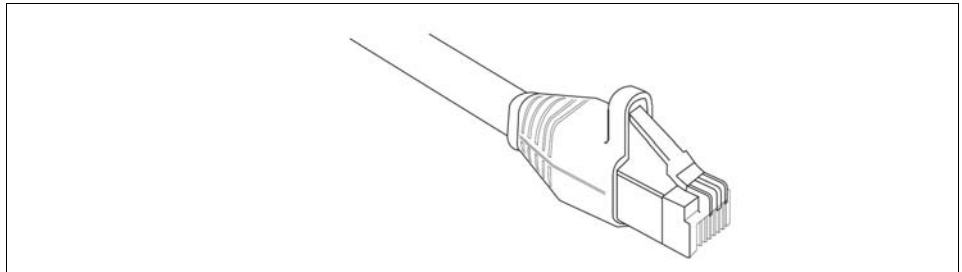
Introduction

TIA/EIA 568 specifies that the UTP connector for data use be an eight pin RJ45 modular connection, which can be either a plug or a jack.

Termination plugs

If a long patch cable is used, it should have an RJ45 plug on both the ends of the cable. Honeywell recommends ordering certified patch cables from a manufacturer that tests all patch cords and offers the cables in yellow or green jackets or with color coded boots that go over the plugs.

Figure 2-8 RJ45 Plug

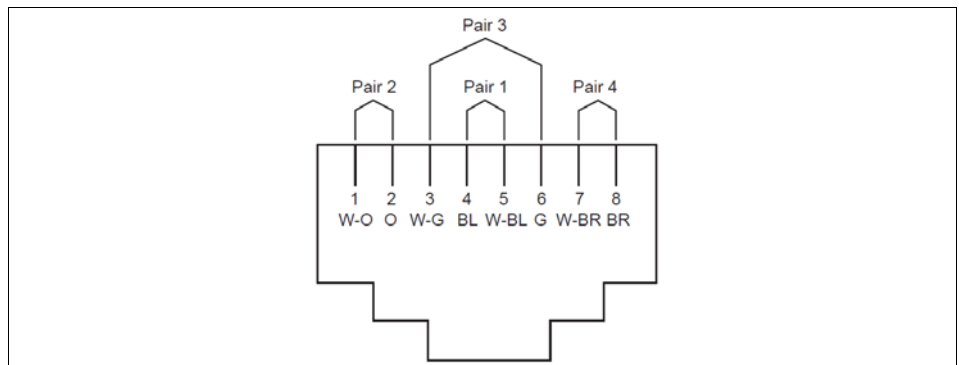


Termination jacks

Usually in structured cable systems, the first point is a service mounted box or flush mounted wall plate with a jack. This jack is attached to the horizontal cable. The other end of the horizontal cable has a screened plug which connects to a jack on the telecom rack.

The other method uses a jack terminated on either end of the horizontal cable. At the location of the network switches, install a patch panel or surface mounted box to house the jacks. At the location of the node, install a jack in a surface mounted box or a flush mounted wall plate. Use the patch cables to connect the jacks to the network switches and to the nodes. Honeywell recommends ordering color coded jacks in both green and yellow and use one patch panel for all the green cables and another patch panel for all the yellow cables. The recommended pinout for terminating the jacks is the 568B pinout.

Figure 2-9 568B Pinout



2.9 Grounding the copper cables

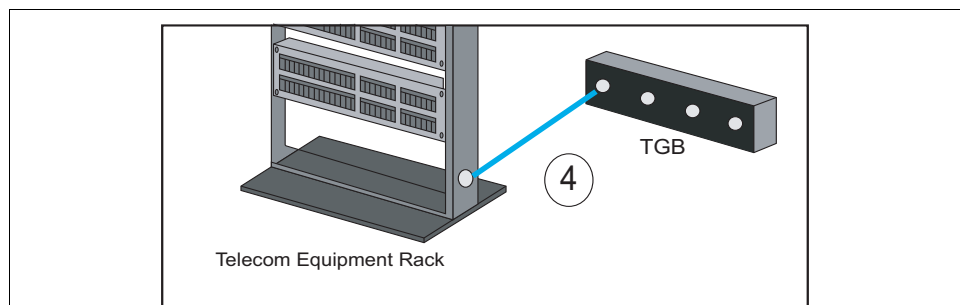
Introduction

This section summarizes grounding procedures for the copper cable plant including the network switches. For more information about grounding copper cables, see *TIA/EIA-568-B.1-2* and *NEC Article 250*.

Telecommunications Grounding Bus Bar (TGB)

Within each rack or telecommunications room, there should be a central grounding point, often called a Telecommunications Grounding Bus bar or TGB. Equipment cabinets should be bonded to this bus bar using a solid copper cable. The equipment in the cabinet are usually grounded through the ground prong on the power plug.

Figure 2-10 Telecommunications Grounding Bus bar



Remote network equipment cabinets

A remote network equipment cabinet is a cabinet that is separate from the main equipment cabinet and contains network equipment connected to the EUCN. Examples include a cabinet containing the FTE switches or a cabinet containing Honeywell's CF9. While it is within the standards to connect these cabinets using copper, unless it is certain that the remote cabinet will be on the same ground bus as the main cabinet, Fiber Optic cable is a better choice. In general, linking these cabinets using copper can be performed safely if:

- The cabinets are in adjacent rooms
- The cabinets are within 60 feet (18.2 meters) of each other
- There are drawings or other firsthand knowledge that the two locations are on the same ground bus
- The jack in the remote cabinet is housed in a non metallic surface mount box

If there is any doubt that the two locations are not on the same ground bus, a ground wire should be installed along with the copper data cable to extend the TGB out to the remote location. Another option is to link the two locations using Fiber Optic. This improves EMI/RFI immunity when compared to extending the TGB.

Importance of using screened cable components

To realize the benefit of a screened cable system, use screened components throughout the installation. This includes cable, jacks, patch panels, and patch cables. See also the installation guidelines that come with each cable product.

Equipment room grounding

In the equipment room, bond the equipment cabinet to the ground, also known as the TGB. After the cabinet is properly grounded, ground the patch panel:

- Attach a 6 AWG ground wire to the grounding lug on the patch panel and attaching the other end of the ground wire to the cabinet, or
- Verify there is a bare metal connection between the mounting rails in the cabinet and the patch panel through the mounting screws.

Bond the shield of the horizontal cable either to the shield of the jack or to a common grounding point on the patch panel. When the jack is terminated at the outlet end, the shield is bonded to the shield of that jack. This ensures a complete ground path from the node outlet jack all the way back to the equipment room ground. Verify only the end in the equipment room is bonded to the TGB.

Importance of using the same grounding system

Short, intra-cabinet connections are assumed to be grounded to the equipment through the cable shield and the equipment chassis ground. It is not expected that equipment within the same cabinet or cabinet row would be on different ground potentials.

Verify the equipment room and the node location are on the same grounding system, and that there is no difference of potential between the two. A difference of potential could result in equipment damage or personal injury. TIA/EIA-568-B.1-2-5.1-4.6 stresses that —At the end of the horizontal node cabling, the voltage measured between the screen and the ground wire of the electrical outlet (used for providing power to the equipment) shall not exceed 1.0 V rms and 1.0 V dc. The cause of any higher voltage should be removed before using the cable.

2.10 Selecting fiber cables for the EUCN

Introduction

Selecting the fiber optic cables is dependent on satisfying installation and environmental requirements without exceeding the maximum optical losses. Consider these factors when selecting fiber optic cable:

- Total fiber optic cable losses
- Cable requirements caused by the desired routing.
- Routing requirements can include direct burial, conduits, trays, raceways, plenums, and so on
- Construction code requirements

This section describes the different characteristics of fiber optic cables that can be used in the EUCN network.

Use of fiber optic patch cables

Fiber optic patch cables are used for connecting the fiber optic patch panel to a piece of equipment or for connecting different strands within a fiber optic patch panel. They are relatively light duty cables usually consisting of 2 fiber optic strands within a duplex PVC jacket with Kevlar yarn added for strength.

Fiber optic cable type

In EUCN, multi-mode (MM) fiber optic cable (CC-FMMX01) is used for data communications.

Fiber optic cable jacket types

Fiber optic cable jackets are available in different armoring configurations, depending on site environmental requirements. Always consider grounding requirements when using armored cable. The following table describes the three main categories of jackets used for fiber optic cable.

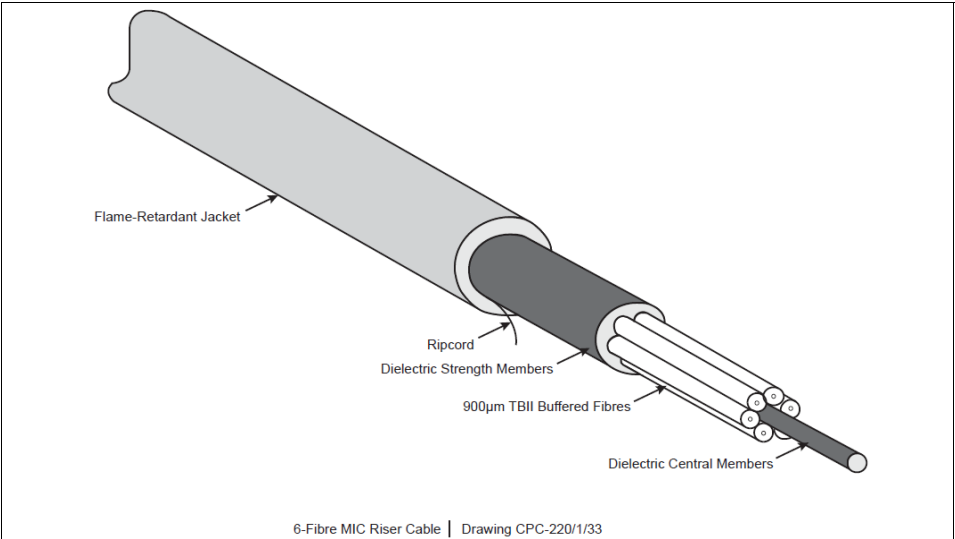
Table 2-5 Jacket types used for Fiber Optic Cable

Jacket Type	Description
Outside plant cable	Intended only for use outside buildings and is not a listed assembly by the NEC for use indoors. Jacket is usually made of polyethylene (PE) and may include metallic armoring. Per article 770.113 an unlisted cable may penetrate a building a maximum of 50 feet (15 meters) before it is required to be terminated, spliced to a listed cable, or enclosed in conduit. Most durable of the three jacket types.
Indoor/outdoor cable	Indoor/Outdoor cabling has a type of jacket that is UL listed for indoor riser and outdoor use, but cannot be used in a plenum airspace. Typically the cabling has a polyvinyl chloride (PVC) jacket with ultra-violet (UV) stabilizers, and includes some type of water blocking compound to enable it to be used outside. Consider a medium duty cable and meets the requirements of NEC Article 770 as long as the building environment is classified as a riser and not a plenum.
Inside plant cable (includes plenum and riser versions)	Inside plant cable can be a substantial assembly, but it is not considered as rugged as an outside plant or indoor/outdoor cable. An inside plant cable jacket is typically either PVC or Teflon, for plenum rated applications, and usually does not contain any water blocking mechanisms.

Fiber optic cable construction

Tight buffered is typically the construction of choice today and is recommended by Honeywell. In this type of cable, each strand has its own jacket extruded over it in the manufacturing process. Typically a water absorbing material is used in place of gel as a water blocking mechanism, making it much easier to prepare the cable for termination.

Figure 2-11 Typical Tight Buffered Fiber Optic Cable (image courtesy of Corning Cable Systems)



Air blown fiber optic cable

Air Blown Fiber (ABF) separates the conventional fiber optic cable into its two components: the outer jacket and the fiber optic strands. The outer jacket component is a tube that contains a number of smaller tubes within the outer jacket. The fiber optic strand component is actually a foam covered bundle of fiber optic strands. The fiber optic bundle is blown along the tubes by either pressurized air or nitrogen.

Honeywell can help determine if this system is appropriate for your site.

Fiber optic connector types

There are a number of standards compliant connector types available. The following table summarizes the uses for each type within the horizontal cable plant.

Table 2-6 Standard Compliant Connector types

Connector Type	Description
SC Connectors	Duplex SC Connector recommended for new fiber optic installations (unless matching an existing type of connector is necessary)
LC Connectors	Smaller connectors used for very high density installations. Used for Level 1 and Level 2 switches. The LC connectors are used for single mode connections to the CF9. The connectors are supported by the Ethernet switches for uplinks and

	CF9 connections.
MT-RJ Connectors	Pre-terminated cable assemblies used for outdoor applications for Duplex multimode connections. Used for CF9 switches. The MT-RJ connector is a multimode fiber connection used by the CF9.

2.11 Routing fiber optic cables

Introduction

Fiber optic cable is the specified media for all inter building connectivity. You can route Fiber optic cable underground (with direct burial cable), through the air (with outdoor aerial cable), or in cable trays and electrical wiring trays. Fiber optic cable is immune to interference from electromagnetic fields or transmissions. Fiber optic cable is safe to route through intrinsically safe areas with no danger of explosion. This section provides recommendations for fiber optic routing.

Fiber optic cable path design

Fault Tolerant Ethernet is built upon a traditional 802.3 CSMA/CD Ethernet architecture. This means that a Level 2 switch is directly connected to a Level 2 backbone switch through the cable plant. The goal of the cable plant design is to route the fiber optic cable in a manner that allows for the greatest redundancy within the existing plant structure.

Separation of A network and B network

Route the A (yellow) and B (green) to the destination using different routes to avoid simultaneously damaging or cutting both the cables.

ATTENTION

The difference in the total length of the routed Link A and Link B cables must be less than 500 meters (1640 feet) to limit the communications delay difference.

Fiber optic cable distance limitations

Physical cable distance is critical to the cable routing design. The maximum distance of an MM fiber optic cable is 2 km (1.24 miles). The distance data is dependent on the physical condition of the cable and the connections.

Fiber optic cable bend radius and pulling tension

The Cable manufacturers publish the minimum bend radius and the pulling tension guidelines for the cables. Follow the guidelines at all times, as over bending a cable can distort the pairs and impede performance. If you use plastic tie-wraps to dress cables, do not over tighten and indent the cable jacket. Consider using Velcro tie-wraps to avoid this problem. The general guidelines for the load bend radius for fiber optic cables are as follows:

- Outside plant fiber optic cable must have a minimum pulling strength of 600 lbs/foot
- Outside plant fiber must have a minimum bend radius of 10 times the cable diameter at rest and 20 times the cable diameter under load.
- Air blown fiber optic cable has its own bend radius requirements related to the tube in to which the fiber bundles are installed in. Contact the cable manufacturer for specifications.

Direct burial hazards

Heavy-duty cables can be buried directly in the ground. Use cable with a durable polyethylene jacket material and an inner layer of steel armor to provide protection from the effects of the following conditions:

- Freezing water
- Heaving of rocks caused by the ground freezing
- Ground disruption because of construction
- Rodents

You can also choose to bury the cable in conduit or polyethylene pipe.

General fiber optic safety guidelines

- All unused fiber ports and interfaces should have their protective caps in place.
- Only trained personnel must handle and install fiber optic cables.

WARNING

When working around equipment with fiber optic interfaces or fiber optic patch panels, never look directly into an interface or port. Doing so may result in permanent eye damage.
Do not handle the actual fiber optic strands as these can break and be lodged in your skin.

2.12 Identifying network cables

Introduction

This section contains guidelines for identifying the network cables.

Labeling all network cables

Before installing data cabling, establish a standard labeling convention for all the network cables. TIA/EIA 606 recommends a scheme for labeling large cable plants. However, for smaller systems, Honeywell recommends you to establish a standard method and convention for labeling all the network cables, patch panels and outlets.

Cable identification examples

Create the same label for both the ends of the cable. The label must include a unique cable identification number, indication of where both the ends of the cable terminate, and an identifier for the FTE network tree. You can also add additional characters to indicate the type of cable or the number of strands. The following table shows cable examples.

Table 2-7 Cable identification examples

Unique cable number	Location of termination point	Location of termination point	FTE network identifier
40-	NCB-	SCB-	Y
40 th cable	North Control Building	South Control Building	Indicates yellow connection
50-	WCB-	ECB-	G
50 th cable	West Control Building	East Control Building	Indicates green connection

2.12 Splicing and connecting fiber optic cables

Cabling design considerations

When planning a system installation, design the cabling to have the minimum number of splices. For example, converting an outdoor cable to an indoor cable when entering a cabinet where the bend radius of the outdoor cable does not fit in the cabinet. Provide sufficient cable length for a splice loop. For a service loop, 30 to 45 centimeters (12 to 18 inches) on each cable end is the usual allowance. Also, when entering the equipment cabinet, sufficient cable length must be allowed for breakout (stripping and fanning out) and termination of the individual fibers.

Cable splice protection

Completed splices cannot withstand tensile forces and must be housed in a strain relief assembly. Moisture entry into the splice can cause degradation of performance. Therefore, the splice enclosure must be sealed, and if necessary, the splice encapsulated to minimize moisture entry.

Cable breakout

Breakout eliminates the outer sheath, leaving the more flexible individual fibers for routing within the cabinet. Tight buffer indoor cables provide strength members with each fiber, eliminating the need to use breakout kits to add strength members for each fiber.

Cables with connectors preinstalled

If the cable length is accurately determined and the cable does not have to be pulled through a conduit, cables can be ordered with connectors installed on the ends. This requires less field skill and installation time. To minimize losses at or near maximum cable lengths, professional installation is required using fusion splices.

Vertical fiber migration consideration

Fiber in vertical installations does not break because of its own weight. However, for vertical runs of 15 meters (49.2 feet), and greater, excess fiber can migrate downward. The crowding of excess fiber at the bottom can cause an increase in attenuation.

You can reduce this downward migration of fiber in vertical runs by placing loops in the cable, approximately 0.3 to 0.5 meter (1 to 1.5 feet) in diameter, at the top, bottom, and at 15-meter (50-feet) intervals.

Securing network cables

Aerial Lashing methods are similar to those used for electrical cables. Most cables are compatible with helical lashing, clamping, or tied mounting.

You must firmly clamp the cable in vertical trays, raceways, or shafts at frequent intervals to support cable weight evenly. Clamping intervals can be as short as one meter (3.3 feet) outdoors to prevent wind slapping and minimize ice loading, or as long as 15 meters (49.2 feet) in interior locations.

2.13 Testing EUCN cables

Introduction

This section provides testing guidelines for installed EUCN cabling. Testing applies only to a structured cable solution, as the test equipment is not designed to test short patch cables in the field. This is another reason that patch cables should not be field assembled and should be purchased from a manufacturer that does 100% testing of their cables.

Additional Information

Refer to the following publications

- Copper cabling: TIA/EIA-568-B.1.11.2 and TIA/EIA-568-B.2-1 Annex B
- Fiber optic cabling: TIA/EIA-568-B.1.11.3, TIA-526-7 and TIA-526-14-A.

Multi-test equipment

TIA/EIA standards define the test parameters for installed data cabling and the performance requirements for the testers used. As a minimum, all the installed horizontal or backbone cables should be tested with one of the new cable test devices designed for both copper and fiber testing. The device tests both fiber and copper using an Autotest procedure for a number of standards based parameters and is sufficient for most installations.

Figure 2-12 Fluke DTX Tester



Basic Test Parameters

A few of the most basic test parameters for copper cable are:

- Wire mapping: This verifies that the cable is terminated properly at each end
- Length: This verifies that the cable does not exceed the maximum length per the standards
- Near End Crosstalk (NEXT): This also verifies that the cable has been installed and terminated properly. Hence, ensuring that the manufactured electrical characteristics of the cable are able to cancel out crosstalk between adjacent cable pairs near the transmitting source.
- Insertion and Return Loss: Insertion loss is a measure of signal loss in the permanent link or channel. Return loss is a measure of the reflected energy caused by impedance variations in the cabling system.
- Propagation delay: Propagation delay is the time it takes for a signal to propagate from one end to the other.

The testers can also be set to test for either Cat 5e or Cat 6 and use the correct test parameters for the installed system.

Test Procedure – Fiber Optic Cables

When using a multi test device to test the copper cable, a tester is connected to each end of the cable under test and the Autotest sequence is started. Usually, the tester prompts to switch the fiber strands partway through the test. This reverses the transmit and receive strands so that the cable is tested in both the directions. When using a loss test set, there is usually no Autotest function and all the measurements are taken manually. The loss test set consists of two units, a light source and a light detector. The two units are connected together in order to be calibrated; then one unit is connected to each end of the fiber. The measurement displayed on the screen is the loss of the cable and the connectors, usually measured in dBs. Typical loss budgets for Cisco interfaces are around 9 dB. However, loss of more than 4 dB should be investigated. TIA allows .75 dB of loss for each connector, plus the loss inherent in the fiber which is usually measured in dBs per kilometer.

Fiber optic cable test equipment

Longer outside plant fiber installations should also be verified with an Optical Time Domain Reflectometer (OTDR). An OTDR is a single device that is connected to the test cable using a “launch cable”, which is just a long fiber jumper. The OTDR transmits a constant light signal and measures the amount of that signal that is scattered inside of the fiber back to the OTDR. Since the OTDR knows the properties of the particular fiber (data input is part of the set up of the OTDR), it can calculate the actual length of the cable and other characteristics from its measurements. If the cable plant is tested with an OTDR, the trace should be saved for future reference. The OTDR visually displays the installed cable as a wave form. It also displays any events or changes to the cable such as splices or kinks, and provides the more accurate length measurement compared to a multi test device.

Test results

On completion of the testing, the test results should be reviewed against the passing levels and provided to the cable plant owner. Also, verify that appropriate information such as jack number and location is included in addition to the pass or fail results. The test results should be saved for future reference.

Section 3 – Cable Fabrication

3.1 Overview

Introduction

This section of the manual provides instructions for fabricating trunk and drop cable assemblies on site with materials supplied in Honeywell connector kits. The cables and drop cable taps are then used to install a Universal Control Network. Only Honeywell-supplied or approved bulk cable is covered by these instructions.

3.2 RG6-Type Coaxial Drop Cable Assembly

Introduction

The following instructions are for installing the Gilbert Datagrade F-type coaxial cable connector on standard Universal Control Network (UCN) quad-shield RG6-type drop cable.

Reference

Gilbert Engineering in Phoenix, Arizona
(602) 245-1050 or 800-528-0199

Tools required

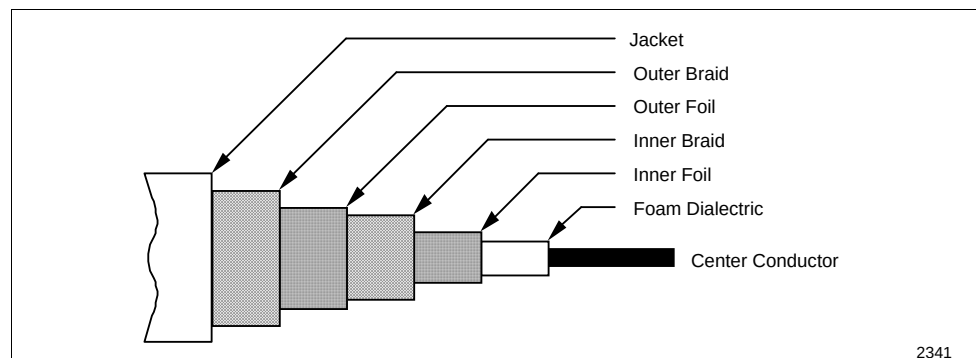
The following tools are required.

- Gilbert Engineering G-CRT-188 Crimp Tool
(Ben Hughes Communications HCT-188)
- Coax cable cutters

Exposed view of RG6-type cable

Figure 3-1 is an exposed view of the RG6-type coaxial cable.

Figure 3-1 Exposed View of RG6-Type Coaxial Cable



3.2 RG6-Type Coaxial Drop Cable Assembly, Continued

RG6-type cable fabrication procedure

Use the procedure in Table 3-1 to fabricate RG6-type drop cables.

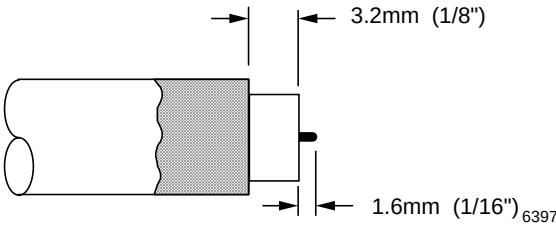
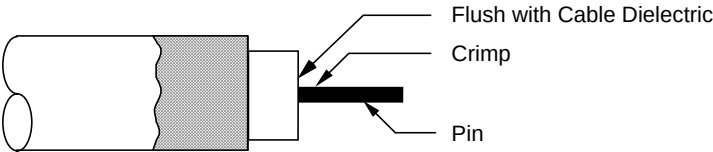
Table 3-1 RG6-Type Cable Fabrication Procedure

Step	Action
1	Cut the cable end at a clean 90° angle.
2	Remove 7.9 mm (5/16") of the jacket. Do not cut the underlying outer braid. 6398
3	Fold the exposed outer braid back over the jacket. 6395
4	Remove the outer foil. Do not cut the inner braid under the foil. Fold the inner braid back over the jacket. Do not disturb the inner foil. 6396
5	Remove 4.8 mm (3/16") of the dielectric and the inner foil. Do not score the center conductor. 6393

3.2 RG6-Type Coaxial Drop Cable Assembly, Continued

RG6-type cable
fabrication procedure,
continued

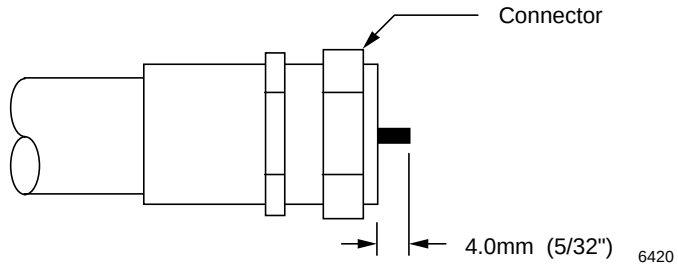
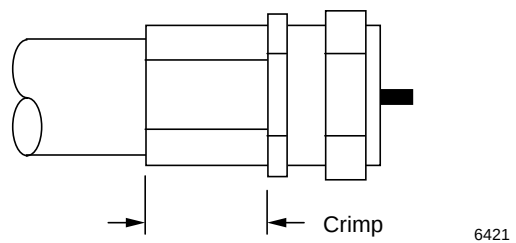
Table 3-1 RG6-Type Cable Fabrication Procedure, Continued

Step	Action
6	Trim the center conductor to 1.6 mm (1/16") beyond the dielectric. Do not distort the circular end of conductor.  Cable preparation is now complete. Next, the separate crimp pin is crimped onto the cable's center conductor. Finally, the connector body will be applied to the cable and crimped. Proceed to steps 7 -11.
7	Push the connector pin over the cable center conductor until flush with the dielectric. Crimp the pin with a 0.054 hexagon crimp (ref: G-CRT-188 crimp tool).  Figure 3-2 is an illustration of the crimp tool.
8	Minimize the thickness of the two folded-back braid layers by straightening and intermingling the strands. This will ease the following application of the connector over the end of the cable and braid.

3.2 RG6-Type Coaxial Drop Cable Assembly, Continued

RG6-type cable
fabrication procedure,
continued

Table 3-1 RG6-Type Cable Fabrication Procedure, Continued

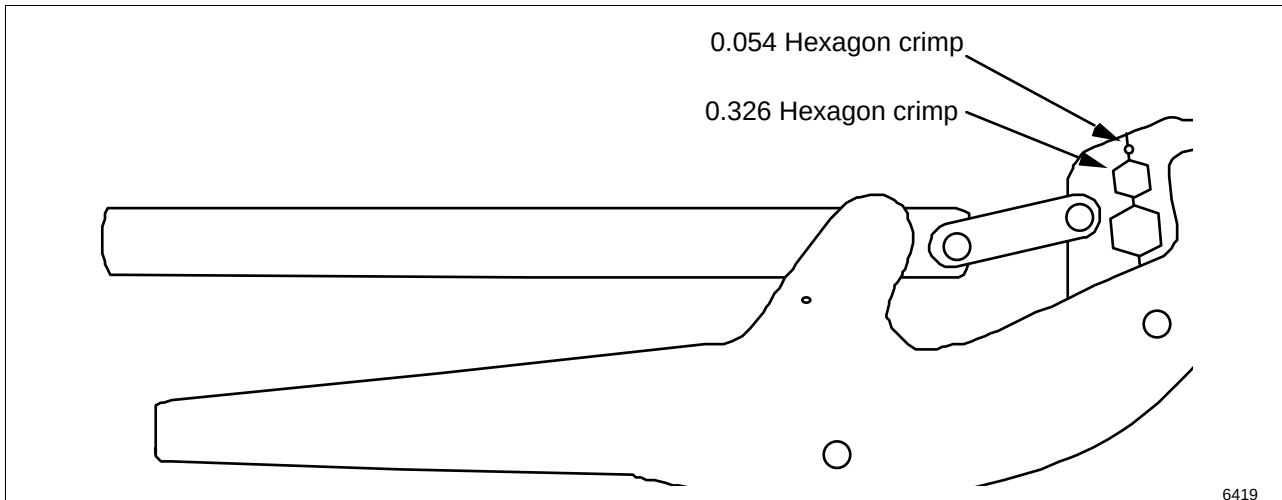
Step	Action
9	Ensure that the inner foil is still wrapped neatly (flush) around the dielectric. Guide the pin into the insulator inside the connector body. Push the connector onto the cable until the pin protrudes approximately 4.0 mm (5/32") beyond the connector's end and a reasonable stop is felt. The braid shield should not be visible beyond the crimp ring. 
10	Crimp the connector with a 0.326 hexagon crimp (Ref: G-CRT-188 crimp tool). 
11	Apply precut adhesive-backed, heat-shrinkable tubing supplied with the connectors in the kit, and an orange (cable A) or brown (cable B) label, also supplied, as illustrated in Figure 3-3. Hand tug on the installed connector. Do not exceed 10 lbs pull. Recheck the pin's protrusion in Step 9 above.

3.2 RG6-Type Coaxial Drop Cable Assembly, Continued

Model G-CRT-188 crimp tool

Figure 3-2 illustrates the model G-CRT-188 crimp tool.

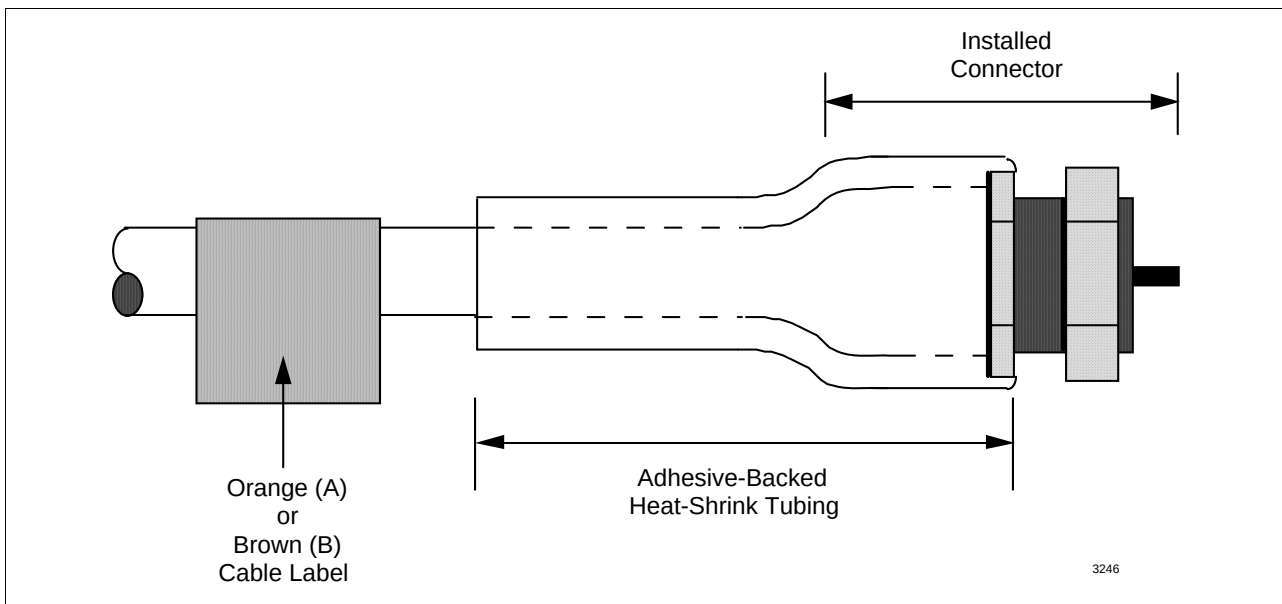
Figure 3-2 Model G-CRT-188 Crimp Tool



Application of cable tubing and label

Figure 3-3 illustrates the application of heat-shrink tubing and the identification label on the cable.

Figure 3-3 Application of Heat-Shrink Tubing and Cable Label



Cable testing

Check the finished cable assemblies for proper electrical performance with procedures found in Section 5. Apply plastic protectors, also supplied, to the ends of the finished cable as illustrated in Figure 3-3. To prevent damage to the protruding pin, the protectors should remain in place until the cable is installed.

3.3 RG11-Type Coaxial Trunk Cable Assembly

Introduction

The following instructions are for installing the Gilbert Datagrade F-type coaxial cable connector on standard Universal Control Network (UCN) quad-shield RG11-type drop cable.

Reference

Gilbert Engineering in Phoenix, Arizona
(602) 245-1050 or 800-528-0199

Tools required

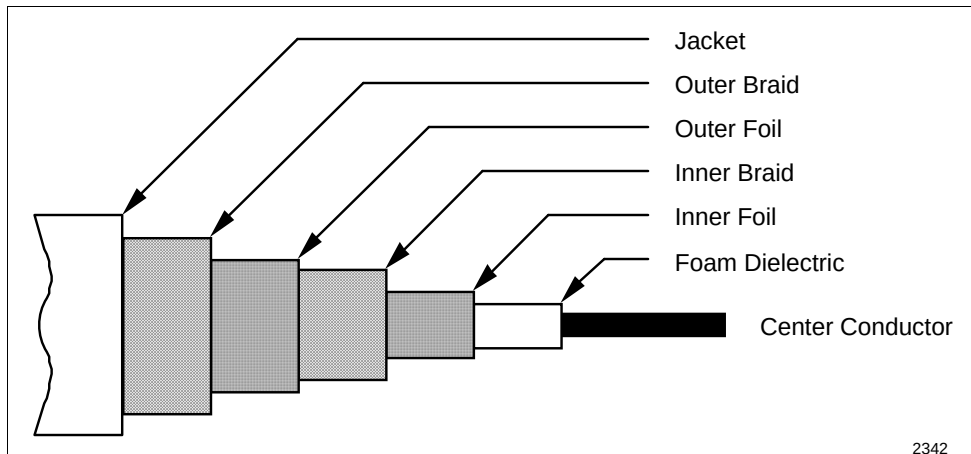
The following tools are required.

- Gilbert Engineering G-CRT-211 Crimp Tool (Ben Hughes Communications HCT-211)
 - Coax cable cutters
-

Exposed view of RG11-type cable

Figure 3-4 is an exposed view of the RG11-type coaxial cable.

Figure 3-4 Exposed View of RG11-Type Coaxial Cable

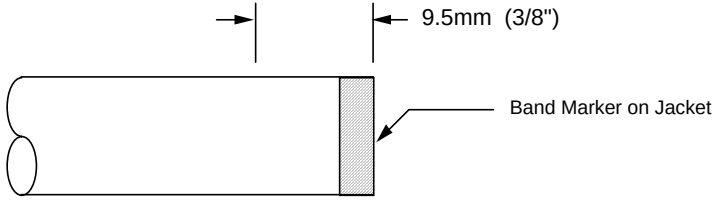
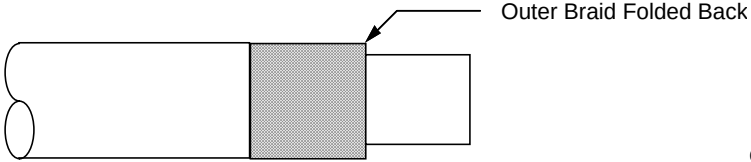
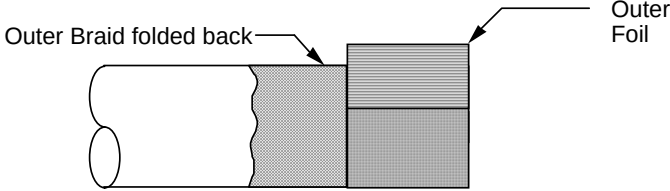


3.3 RG11-Type Coaxial Trunk Cable Assembly, Continued

RG11-type cable fabrication procedure

Use the procedure in Table 3-2 to fabricate RG11-type coaxial trunk cables.

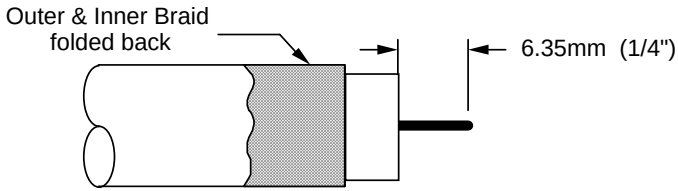
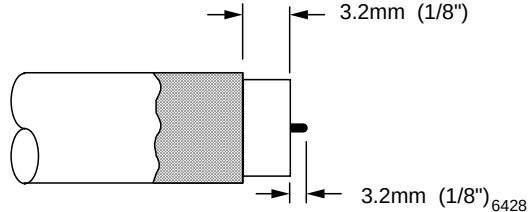
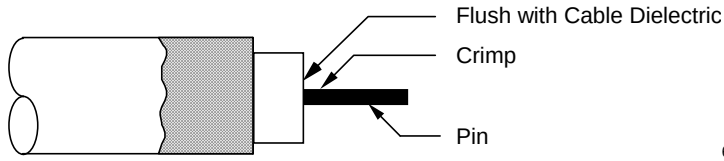
Table 3-2 RG11-Type Cable Fabrication Procedure

Step	Action
1	Cut the cable end at a clean 90° angle. Both ends of the cable must be cut at the periodic (2.6-meter (8.5 foot) interval) band marks on the trunk cable outer jacket only.
2	Remove 9.5 mm (3/8") of the jacket. Do not cut the underlying outer braid.  6424
3	Fold the exposed outer braid back over the jacket.  6425
4	Remove the outer foil. Do not cut the inner braid under the foil. Fold the inner braid back over the jacket. Do not disturb the inner foil.  6426

3.3 RG11-Type Coaxial Trunk Cable Assembly, Continued

RG11-type cable
fabrication procedure,
continued

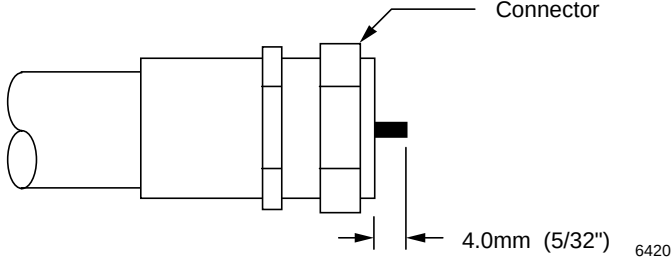
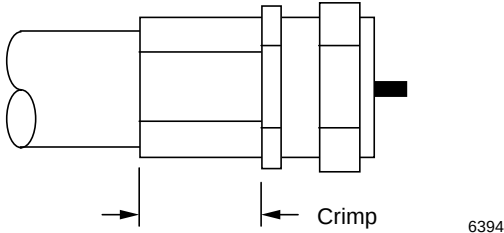
Table 3-2 RG11-Type Cable Fabrication Procedure, Continued

Step	Action
5	Remove 6.35 mm (1/4") of the dielectric and the inner foil. Do not score the center conductor.  6427
6	Trim the center conductor to 3.2 mm (1/8") beyond the dielectric. Do not distort the circular end of conductor.  Cable preparation is now complete. Next, the separate crimp pin is crimped onto the cable's center conductor. Finally, the connector body will be applied to the cable and crimped. Proceed to steps 7 -11.
7	Push the connector pin over the cable center conductor until flush with the dielectric. Crimp the pin with a 0.10 hexagon crimp (reference: G-CRT-211 crimp tool).  Figure 3-5 is an illustration of the crimp tool.
8	Minimize the thickness of the two folded-back braid layers by straightening and intermingling the strands. This will ease the following application of the connector over the end of the cable and braid.

3.3 RG11-Type Coaxial Trunk Cable Assembly, Continued

RG11 cable
fabrication procedure,
continued

Table 3-2 RG11-Type Cable Fabrication Procedure, Continued

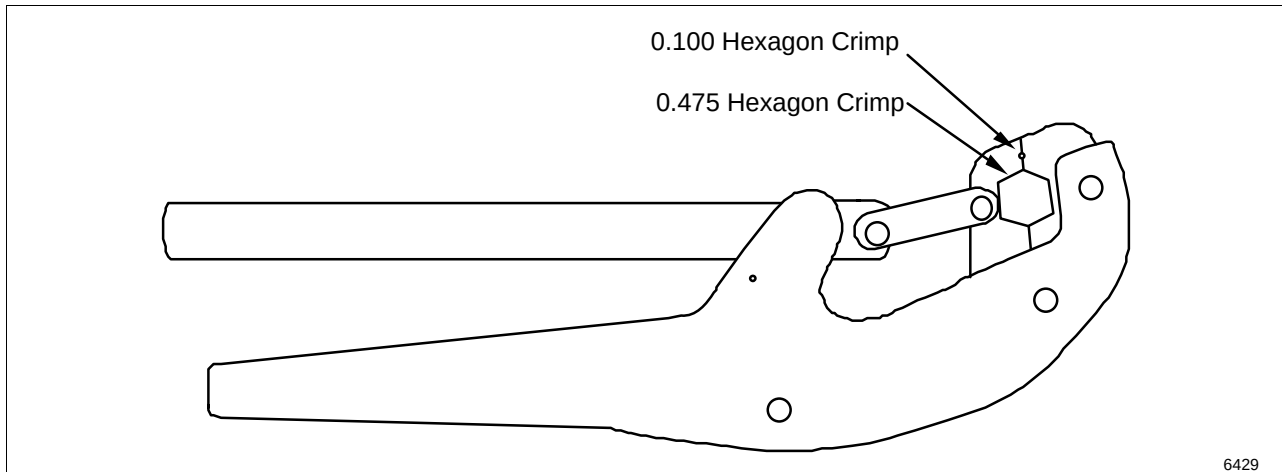
Step	Action
9	Ensure that the inner foil is still wrapped neatly (flush) around the dielectric. Guide the pin into the insulator inside the connector body. Push the connector onto the cable until the pin protrudes approximately 4.0 mm (5/32") beyond the connector's end and a reasonable stop is felt. The braid shield should not be visible beyond the crimp ring. 
10	Crimp the connector with a 0.475 hexagon crimp (reference: G-CRT-211 crimp tool). 
11	Apply precut adhesive-backed, heat-shrinkable tubing supplied with the connectors in the kit, and an orange (cable A) or brown (cable B) label, also supplied, as illustrated in Figure C-3. Hand tug on the installed connector. Do not exceed 10 lbs pull. Recheck the pin's protrusion in Step 9 above.

3.3 RG11-Type Coaxial Trunk Cable Assembly, Continued

Model G-CRT-211 crimp tool

Figure 3-5 illustrates the model G-CRT-211 crimp tool.

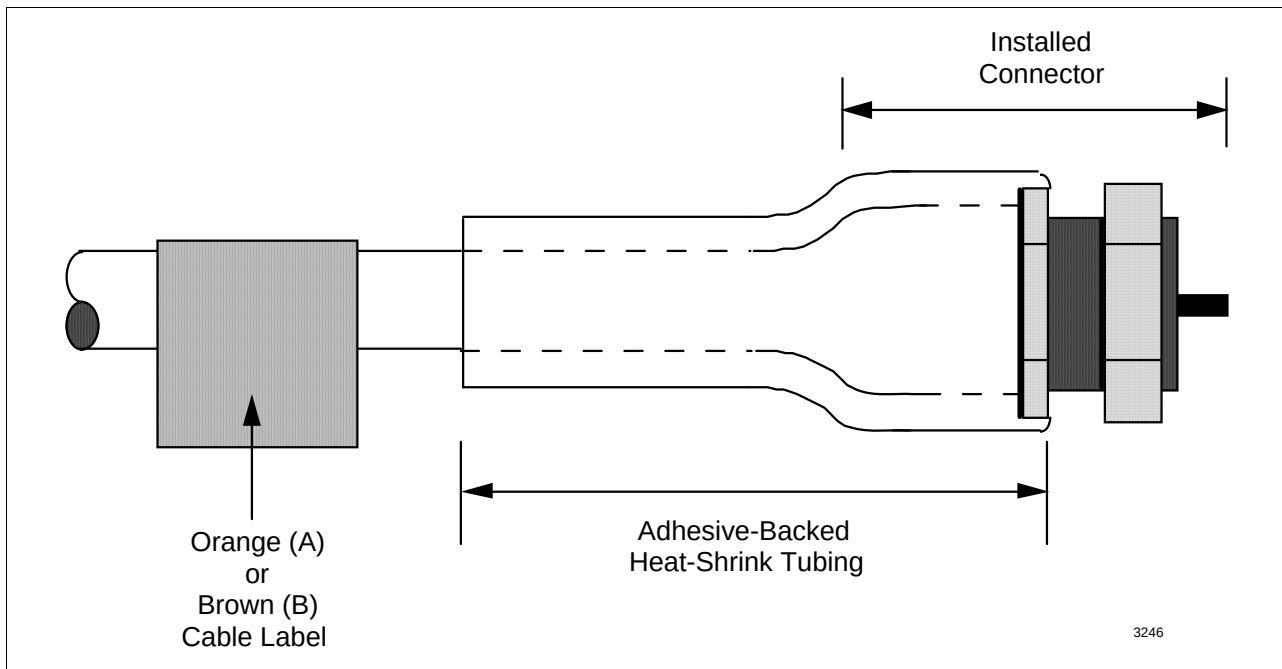
Figure 3-5 Model G-CRT-211 Crimp Tool



Application of cable tubing and label

Figure C-6 illustrates the application of heat-shrink tubing and the identification label on the RG11-type cable.

Figure 3-6 Application of Heat-Shrink Tubing and Cable Label



Cable testing

Check the finished cable assemblies for proper electrical performance with procedures found in Section 5. Apply plastic protectors, also supplied, to the ends of the finished cable as illustrated in Figure C-3. To prevent damage to the protruding pin, the protectors should remain in place until the cable is installed.

Section 4 – Universal Control Network Extender Installation

4.1 Overview

Introduction

This section provides overview information for configuration and installation of the Universal Control Network Extender.

4.2 Description

Purpose

The Universal Control Network two-kilometer fiber optic Extender (UCNE) is used to:

- increase the end-to-end distance of a single UCN without resorting to special, wide-diameter coaxial trunk cables.
- increase the UCN signal quality at the ends of the UCN that would otherwise be connected by long coaxial trunk cables.
- implement certain UCN topologies, such as a “star configuration,” that is not possible by using coaxial trunk cables.
- provide ground isolation, when necessary, between UCN trunk segments.

Each UCNE provides coax-to-fiber optic and fiber optic-to-coax cable conversion for the UCN signals.

4.3 Universal Control Network Topologies

Implementation

UCNEs are installed in pairs at each end of a redundant fiber optic link; that is, a UCNE-A and UCNE-B at one end of the fiber optic link and a UCNE-A and UCNE-B at the other end of the fiber optic link. Therefore, a total of four UCNEs are required to implement one redundant fiber optic link.

The UCNE, at each end of the fiber optic link, never connects directly to the UCN coaxial trunk, but instead connects to the drop port of a UCN tap on that coaxial segment.

Standard coaxial UCN drop cables (RG6-type) are used to connect the UCNEs to the UCN tap drop ports.

Introduction

There are two classes of UCN topology when using UCN fiber optic Extenders (UCNEs) to link coaxial UCN trunk segments.

- Serial
- Star

Serial UCN topologies

The simplest “serial” UCN topology consists of two UCN coaxial trunk segments that are interconnected by a single fiber optic link. Figure 4-1 illustrates the “one series link” configuration.

Another serial UCN topology consists of three UCN coaxial trunk segments that are interconnected by two fiber optic links. Figure 4-2 illustrates the “two series link” configuration.

Each UCN redundant fiber optic link requires two UCNEs (UCNE-A and UCNE-B) at each end of the link, or a total of four UCNEs. Two fiber optic links, as illustrated in Figure 4-2, would require eight UCNEs.

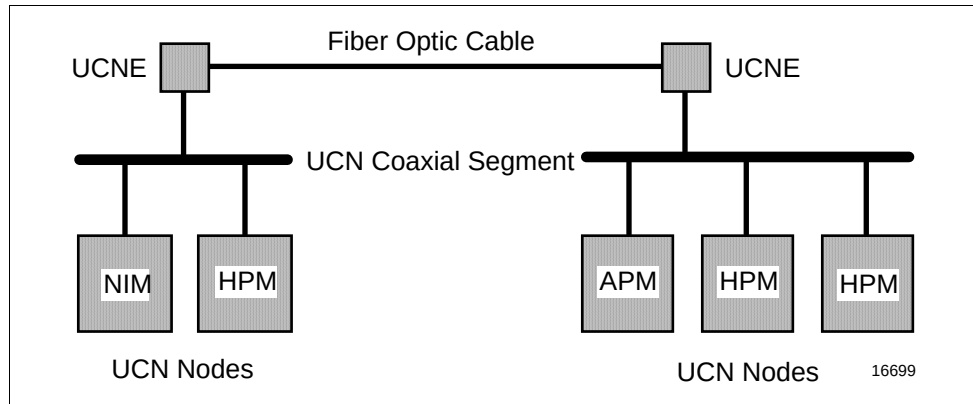
The maximum number of series-connected fiber optic links allowed in a physical UCN is two. The use of three or more fiber optic links in a serial topology is prohibited, because the delay imposed by the additional UCNE(s) and fiber optic cable could violate the UCN’s slot time setting and cause UCN communication errors.

4.3 Universal Control Network Topologies, Continued

Serial UCN topology 2 coaxial segments

Figure 4-1 illustrates a typical serial UCN topology with two UCN coaxial segments and one fiber optic link. Only network “A” is shown. Network “B” is not shown.

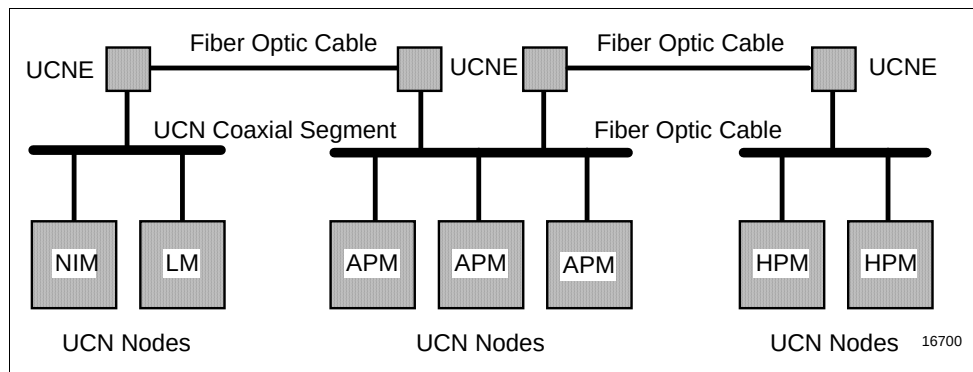
Figure 4-1 Typical Serial UCN Topology – Two Coaxial Segments



Serial UCN topology 3 coaxial segments

Figure 4-2 illustrates a typical serial UCN topology with three UCN coaxial segments and two fiber optic links. Only network “A” is shown. Network “B” is not shown.

Figure 4-2 Typical Serial UCN Topology – Three Coaxial Segments



4.3 Universal Control Network Topologies, Continued

Star UCN topology

A “star” UCN topology consists of a central UCN coaxial cable segment with peripheral UCN coaxial cable segments connected to the central UCN coaxial cable segment through UCNEs. Figure 4-3 illustrates a typical star UCN “star” configuration example.

UCNEs must not be used with passive star couplers because passive couplers have excessive optical power loss. Instead, star topology is implemented by using multiple drop ports on a central UCN coaxial segment to branch out through multiple UCN fiber optic links to other UCN coaxial segments. An example of their use is a central control room that contains the NIM(s) with multiple radial UCN fiber optic links (using UCNEs) to remote UCN coaxial segments that accommodate other UCN nodes. Figure 4-3 illustrates four radial fiber optic links to four remote sites.

Only one of the redundant UCN cables, A or B, is shown in the figure for simplification.

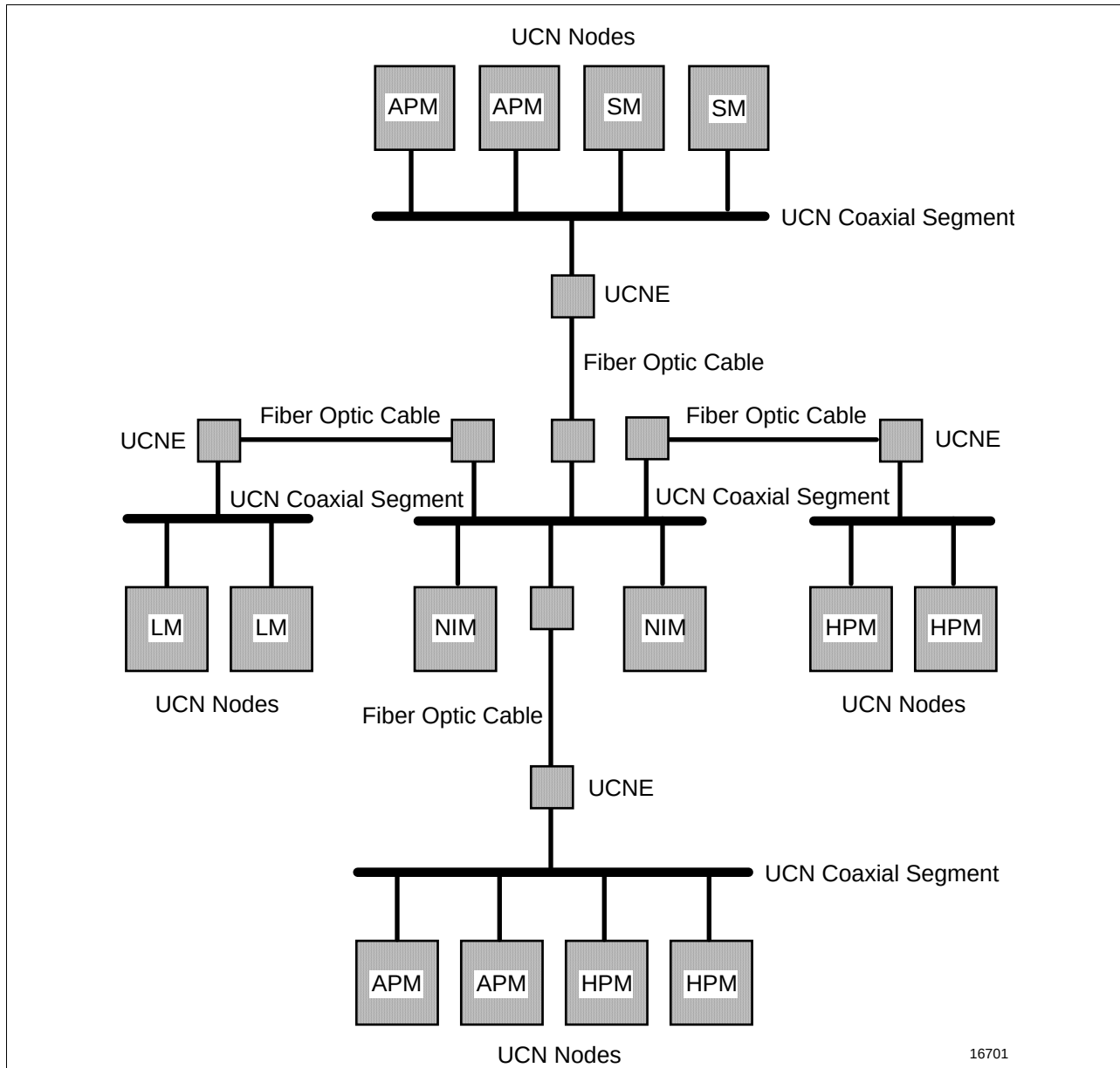
It is preferable that the NIM(s) is at the center of the “star,” but it is not a requirement, just a recommendation, to minimize the “loss of view” in the unlikely event that a redundant fiber optic link fails completely.

4.3 Universal Control Network Topologies, Continued

Serial UCN topology

Figure 4-3 illustrates a typical star UCN topology with five UCN coaxial segments. Only network “A” is shown. Network “B” is not shown.

Figure 4-3 Typical Star UCN Topology



4.4 UCNE Configuration Rules

Configuration rules

Regardless of the UCN topology, the following configuration requirements must be met.

ATTENTION

ATTENTION—UCN node addresses must be assigned to maximize normal token passing from a higher address to the next lower address within the same coaxial UCN, and to minimize that process across fiber-optic segments (through UCNEs) to minimize the overall token rotation time and to prevent UCNE and cable-fault recovery problems. This can be accomplished by using consecutive node addresses within the same coaxial UCN as much as possible.

UCN node addresses 1-8 are reserved for NIMs, with the lowest address being used first. Addressing for nodes other than NIMs begins at 9. Above 9, no odd-numbered addresses can be skipped. This optimizes the performance of the software that controls UCN cable selection.

1. The total length of the UCN (network A or B) between any two nodes, including all series-connected coaxial trunk cable segments and fiber optic links, but not including drop cable lengths, must not exceed 2 kilometers. Since any two UCN nodes can potentially communicate with each other, the restriction is determined by the two nodes that are farthest apart from each other along either UCN cable A or UCN cable B and separated by a combination of UCN coaxial trunk cable and UCN fiber optic cable. Violation of this rule could result in UCN errors that are due to communication delays that exceed the UCN slot time. UCN slot time is the maximum time that any UCN node waits for a response from any other UCN node on the network and is affected by cable and UCNE delays.
 2. Related to rule 1 above, the difference between the total length of UCN network A and UCN network B must be no greater than 500 meters. That is, the UCN A cable and UCN B cable communication paths between the two farthest apart UCN nodes must not be different by more than 500 meters. Violation of this rule could result in UCN errors that are due to communication delays that exceed the UCN slot time.
 3. Related to rule 1 above, the length of fiber optic cable that connects any two UCNEs must not be longer than 2 kilometers, but there is no minimum length requirement. Violation of this rule could result in UCN communications errors because the optical signal loss budget would be exceeded for the interconnected UCNEs (see the following rule).
-

4.4 UCNE Configuration Rules, Continued

4. Related to rule 3 above, the optical signal loss budget for a fiber optic cable connection between two interconnected UCNEs is 10 dB maximum and must not be exceeded. This includes up to 2 kilometers of fiber optic cable that is specified in Honeywell drawing 51190918 that has a maximum loss of 4 dB/km, two ST-type connectors (one at each end of the fiber optic cable) with a maximum loss of 1 dB per connector, and two cable splices with a maximum loss of 0.5 dB per splice. Violation of this rule could result in UCN communications errors because the optical signal loss budget could be exceeded for the interconnected UCNEs.
 5. The UCN coaxial cable configuration rules found in other sections of the manual also apply for each separate coaxial trunk segment of the network.
-

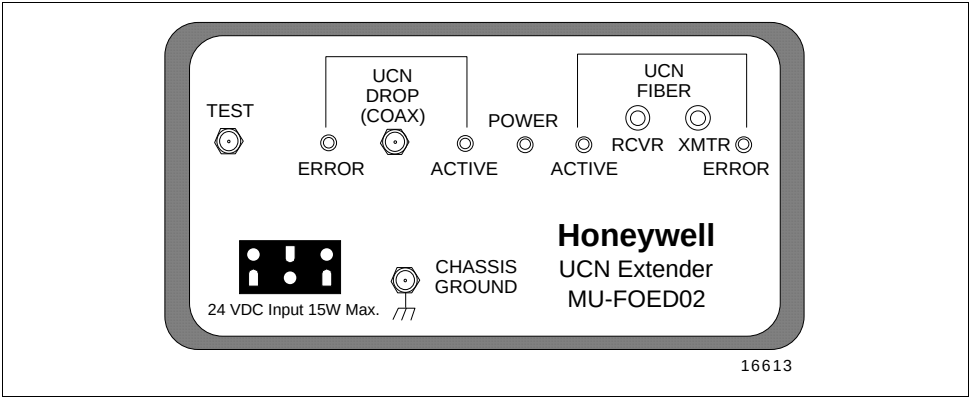
4.5 UCNE Model Numbers

Model numbers	The UCNE model numbers that support the maximum 2-kilometer fiber optic cable distance are: • MU-FOED02 A pair of 18-36 Vdc powered UCNEs • MU-FOEA02 A pair of 85-264 Vac, 47-63 Hz powered UCNEs
DC-powered UCNE	The dc-powered UCNE is used where a PM, APM, or HPM Power System's 24 Vdc is available. Figure 4-4 is an illustration of a dc-powered UCNE.
AC-powered UCNE	The ac-powered UCNE is typically used at UCN locations where a PM, APM, or HPM Power System's 24 Vdc is not available, such as in cabinets or consoles that house a NIM, LM, or SM. However, the ac-powered UCNE should be powered by UPS-backed ac power if it is expected to continue supporting UCN communications during an ac line power outage. Figure 4-5 is an illustration of an ac-powered UCNE.
UCNE independence	UCNEs must be installed independently from each other such that a failed UCNE in a redundant pair (UCNE-A and UCNE-B) can be replaced while the other UCNE continues operating. This is accomplished using the standard mounting kits that are described in subsection 4.6.

4.5 UCNE Model Numbers, Continued

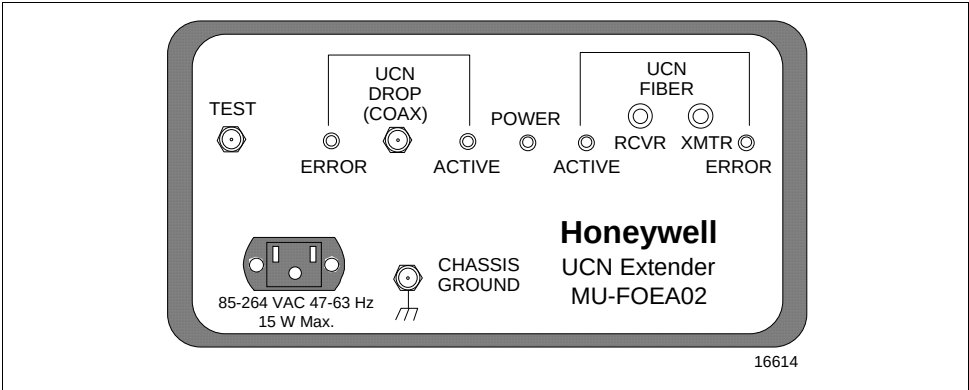
DC-powered UCNE Figure 4-4 illustrates the front panel of a dc-powered UCNE.

Figure 4-4 DC-Powered UCNE



AC-powered UCNE Figure 4-5 illustrates the front panel of an ac-powered UCNE.

Figure 4-5 AC-Powered UCNE



4.6 UCNE Mounting Kits

4.6.1 PM, APM, and HPM Cabinets



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Mounting location

The standard mounting location for a redundant pair of UCNEs is in model MU-FOMK01 mounting kit hardware. The kit's enclosure is installed in a Left or Right 7-slot card file position in the card file side of a PM, APM, or HPM cabinet. Typically, the kit is installed after first mounting all the other PM, APM, or HPM card files that connect to the UCN, and in a cabinet that contains UCN taps because UCNEs connect to the UCN through drop cables to the UCN taps.

Mounting kits MU-FOMK01 and MU-FOMK11

The model MU-FOMK01 mounting kit provides for vertical stacking of up to four UCNEs as illustrated by the layout in Figure 4-6. A redundant pair of dc-powered UCNEs (model MU-FOED02) will occupy vertical slots 1 and 3 as illustrated in Figure 4-6. Table 4-1 lists the parts that are included in the model MU-FOMK01 mounting kit.

The model MU-FOMK11 mounting kit provides the hardware to accommodate a second redundant pair of dc-powered UCNEs for use in vertical slots 2 and 4 for an optional second UCN fiber optic link. The second pair of UCNEs must be ordered separately by ordering another model MU-FOED02. Table 4-2 lists the parts that are included in the model MU-FOMK11 mounting kit.

Each of the four vertical mounting slots includes a slide-out mounting tray that attaches to the bottom of a UCNE by using four screws that are provided in the mounting kit.

Mounting positions 1 and 2 are identified by orange-colored "UCN-A" labels, and mounting positions 3 and 4 are identified by brown-colored "UCN-B" labels.

Mounting kits model MU-FOMK01 and MU-FOMK11 include two 5-meter power cables (51201397-002) to provide power to the dc-powered UCNEs. If longer power cables are needed, they must be ordered separately.

4.6.1 PM, APM, and HPM Cabinets, continued

UCN tap drop cables A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect a pair of UCNEs to a pair of UCN taps. The drop cables are not supplied with the UCNE mounting kits and must be ordered separately.

UCN taps If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

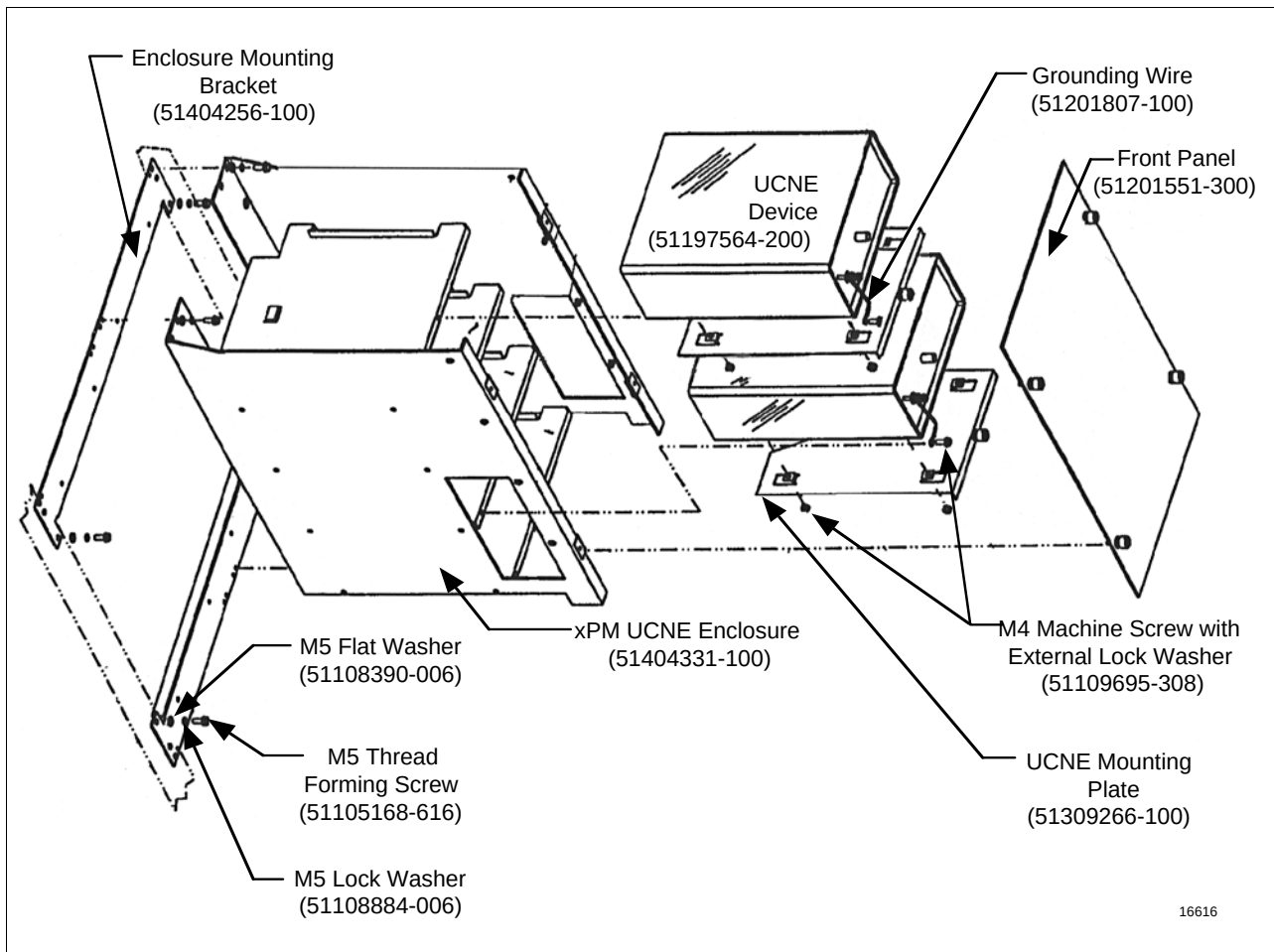
MU-NTAP02	Two 2-drop taps with mounting/grounding hardware
MU-NTAP04	Two 4-drop taps with mounting/grounding hardware
MU-NTAP08	Two 8-drop taps with mounting/grounding hardware

4.6.1 PM, APM, and HPM Cabinets, continued

Pictorial view of model MU-FOMK01 kit

Figure 4-6 is a pictorial view of the model MU-FOMK01 mounting kit. The UCNEs are not included in the kit.

Figure 4-6 Pictorial View of Model MU-FOMK01 Mounting Kit



4.6.1 PM, APM, and HPM Cabinets, continued

Model MU-FOMK01 mounting kit parts

Table 4-1 is the list of parts in the model MU-FOMK01 mounting kit. The pair of UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOED02.

Table 4-1 Model MU-FOMK01 Parts List

Description	Part Number	Quantity
UCNE Enclosure	51404331-100	1
Power Cable – 2 meters (6.5 feet)	51201397-002	2
UCNE Mounting Tray (attaches to UCNE)	51309266-100	2
Grounding Wire	51201807-100	2
Front Panel	51201551-300	1
Enclosure Mounting Bracket (mounts enclosure)	51404256-100	2
Cable Tie	51190879-001	8
M4 Machine Screw	51109695-308	10
M5 Thread Forming Screw	51195168-616	4
M5 Flat Washer	51108390-006	10
M5 Lock Washer	51108884-006	10

Model MU-FOMK11 mounting kit parts

Table 4-2 is the list of parts in the model MU-FOMK11 mounting kit. The pair of UCNEs that are mounted in the UCNE enclosure must be ordered separately by model MU-FOED02.

Table 4-2 Model MU-FOMK11 Parts List

Description	Part Number	Quantity
UCNE Mounting Tray	51309266-100	2
Power Cable – 2 meters (6.5 feet)	51201397-002	2
M4 Machine Screw	51109695-308	10
Grounding Wire	51201807-100	2

4.6.2 LCN Cabinets and Consoles

ATTENTION

ATTENTION—The information that follows is supplemental to the PM/APM/HPM cabinet information. Detailed LCN cabinet/console information can be found in the *Local Control Network Planning* and *Local Control Network Installation* manuals.

Mounting location

The standard mounting location for a redundant pair of UCNEs in LCN cabinets or consoles is in the UCNE enclosures that are included in the following mounting kits.

MP-FOMK02	LCN Cabinet 120 Vac UCNE Mounting Kit
MP-FOMK03	LCN Cabinet 240 Vac UCNE Mounting Kit
MP-FOMK04	Classic Console 120 Vac UCNE Mounting Kit
MP-FOMK05	Classic Console 240 Vac UCNE Mounting Kit
MP-FOMK06-100	Z-Console Upper Pod UCNE Mounting Kit
MP-FOMK06-200	Z-Console Lower Pod UCNE Mounting Kit

The LCN UCNE enclosures can also be installed in LM and SM cabinets. The model MP-FOMK02 and MP-FOMK03 mounting kits are recommended for that purpose.

Model MP-FOMK06 mounting kit

The model MP-FOMK06 mounting kits do not include a pair of replacement leg covers which are required when adding a UCNE enclosure to Z-console furniture. They must be ordered separately. Two factors must be considered before ordering, the height of the console furniture (high, medium, or low) and whether the UCNE enclosure's equipment pod will be the only (upper) pod or a second (lower) pod.

4.6.2 LCN Cabinets and Consoles, continued

Kit descriptions

For each of the kits, a redundant pair of ac-powered UCNEs (model MU-FOEA02), which must be ordered separately, are mounted side by side in a metal enclosure (module). Viewing the rear of the enclosure, the UCNE on the right side of the enclosure is connected to the UCN-A cable and the UCNE on the left side of the enclosure is connected to the UCN-B cable.

Labels on the enclosure identify the UCN-A and UCN-B mounting locations. The labels are color-coded orange and brown, respectively, to agree with the standard UCN redundant cable color-coding scheme. Each mounting kit includes two ac power cables to power the pair of ac-powered UCNEs that must be installed in the enclosure. If redundant ac power sources are available, it is recommended that each UCNE in a redundant pair be connected to a different ac power source to maintain operation of at least one UCNE if a power source should fail. An Uninterruptible Power System (UPS) is recommended to maintain ac power to the UCNEs during ac line outages.

UCN tap drop cables

A pair of UCN drop cables of appropriate length (model MU-NKDxxx where xxx = length in meters) is required to connect the pair of UCNEs to a pair of UCN taps. The drop cables are not supplied with the LCN UCNE mounting kit and must be ordered separately.

UCN taps

If sufficient UCN tap drop ports are not available when expanding a UCN with UCNEs, then either the number of ports per tap must be increased, or the number of taps in the existing UCN coaxial segments (A and B) must be increased. UCN taps can be ordered by model number as follows.

MU-NTAP02	Two 2-drop taps with mounting/grounding hardware
MU-NTAP04	Two 4-drop taps with mounting/grounding hardware
MU-NTAP08	Two 8-drop taps with mounting/grounding hardware

4.7 Grounding

Proper grounding required

The chassis ground stud at the front of the UCNE must be connected to Safety Ground, similar to the grounding of a UCN tap.

A #10 AWG ground wire is provided in all the UCNE mounting kits to connect the UCNE's front panel ground stud to ground on the metal enclosure in which the UCNEs are installed.

The eventual connection to Safety Ground depends on proper grounding of the metal enclosure. As required when installing a UCN tap, the installer must confirm that there is electrical continuity between the UCNE's front panel grounding stud and the cabinet Safety Ground bus bar after the installation is complete. The Safety Ground bus bar is typically located at the bottom of the cabinet.

4.8 Fiber Optic Cable

Recommended cable	The multi-node fiber optic cables recommended for use with a UCNE are the same as the 62.5-micron fiber optic cables that are recommended for the Honeywell I/O Link Extender in the PM, APM, or HPM. Honeywell drawings 51190918 (indoor) and 51190919 (outdoor) define the cables. The cable has 4 dB/km maximum optical attenuation according to the signal loss budget requirements of subsection 4.4. Other fiber optic cable types, such as the 100-micron fiber optic cable, have not been tested with the UCNE by Honeywell and are not recommended. The UCNE is not compatible with single-mode fiber optic cable.
Proper cable construction	UCN indoor fiber optic cable assemblies should be fabricated using the information in Honeywell drawing 51195498. The RCVR and XMTR fibers at each end of the cable should be labeled as such to agree with the connector labeling on the UCNEs.
Cable composition	A UCN fiber optic cable link consists of two transmit-receive pairs of fiber optic (UCN-A and UCN-B). If possible, cables that have high installation costs, such as cables routed in conduit or buried in the ground should have spare fiber optic.
Cable reference	Refer to the <i>Universal Control Network Planning</i> and <i>Local Control Network Planning</i> manuals for fiber optic cable information on the following subjects. • Cable routing • Cable burial • Cable aerial lashing • Cable vertical installations • Indoor cable requirements • Loose buffered cable • Optic fiber losses • Optic fiber signal loss budget calculation • Cable installation • Allowable indoor cable bend radius • Cable construction selection • Cable splices and connections • Power level measurement

4.9 UCNE Installation

4.9.1 Preinstallation Overview

Introduction

This subsection provides the procedures for the installation of UCNE mounting kits in a PM, APM, or HPM cabinet. It does not cover the installation of UCNE mounting kits in an LCN cabinet or console, or in a LM or SM cabinet. That information is found in the *Local Control Network Installation* manual.

Preparation

Before beginning a UCNE installation, be sure you are adequately prepared. Consider the following.

- Are the required number of unused UCN drop tap ports available on each existing UCN coaxial segment, A and B? One port on the A and B segments is needed for each UCN Extender; therefore, two UCNEs will be used.
 - If the required number of usable UCN drop tap ports are not available, have additional taps been ordered to replace existing taps and expand the number of unused ports? If replacement of existing taps is not appropriate, have additional taps been ordered so that they can be added to the trunk cables? When a tap is added to the trunk cable, an additional coaxial cable of the appropriate length is required and the trunks loading factor must be recalculated. Refer to the *Universal Control Network Planning* manual.
 - Have the required number of UCN trunk cables (RG11-type) and UCN taps been ordered to support the coaxial cable segment(s) that will be added to the UCN?
 - Have the required number of drop cables (RG6-type) been ordered to connect the UCNs to the taps and support the coaxial cable segment(s) that will be added to the UCN?
 - Does the Power System have enough unused power connectors to accommodate the UCNE power cables (a minimum of two are required to support a redundant pair of UCNEs)? If the number of unused power connectors is inadequate, the Power System may have to be upgraded to provide additional backpanel connectors (upgrade kit model MU/MC-ZPSR04), or another Power System may be required.
 - Has a contractor been contacted to fabricate and install the fiber optic cables?
-

4.9.2 Installation Procedure Overview

Introduction

Proper installation and modification, if necessary, of the UCN coaxial cable segments must be verified before beginning the installation of the UCNE hardware and the fiber optic link(s).

Coaxial cable segments

Use the rules and procedures in Sections 2 and 3 to fabricate, install, and modify, if necessary, the UCN coaxial cable segments. Relative information can also be found in the *Universal Control Network Planning* manual.

Use the information and procedures in Section 5 to test and verify the coaxial cable segments.

UCNE hardware cabinet mounting

The UCNE hardware is mounted in the card file-side of the PM, APM, or HPM cabinet and occupies half of a card file location. The UCNE enclosure is the same physical height as an HPM 7-Slot card file, but approximately 1 inch wider, and mounts on two mounting brackets in the same way as the HPM 7-Slot card file. The mounting brackets will not accommodate two UCNE enclosures. An enclosure can be mounted in either the left or right position on the mounting brackets, but two UCNE enclosures cannot be mounted side by side on the same mounting brackets.

UCNE enclosure description

A maximum of four UCNEs can be mounted in the UCNE enclosure on four removable trays, although the typical requirement is for only two UCNEs. Each UCNE is secured to its tray by four screws. The tray is secured to its enclosure mounting shelf by a captive screw. After installation of the UCNEs and cabling is complete, a front panel is installed on the enclosure.

UCNE enclosure installation procedure overview

The model MU-FOMK01 mounting kit provides the UCNE enclosure and the required installation hardware for two UCNEs. Two UCNEs are provided in the model MU-FOED02 kit. Both kits are needed to complete the installation of the first two UCNEs.

If four UCNEs are required for the installation, the model MU-FOMK11 mounting kit provides the hardware necessary to mount the two additional UCNEs that are provided by another model MU-FOED02 kit. Both kits are needed to add the optional third and fourth UCNEs to the UCN enclosure.

4.9.2 Installation Procedure Overview, continued

ATTENTION

ATTENTION—Installing a UCNE enclosure between two existing card files, above and below the card file location that is chosen, is difficult without a socket drive with a long extension. A 1/4-inch drive, a minimum-length 14-inch extension, and an 8-mm (3/8-inch) socket is recommended to have access to the UCNE enclosure attachment screws.

4.9.3 UCNE Enclosure Installation Procedure

UCNE enclosure installation procedure

Use the procedure in Table 4-3 to install a UCNE enclosure and UCNEs.

Table 4-3 UCNE Enclosure Installation Procedure

Step	Action
1	Decide on an available card file position in the cabinet to install the UCNE enclosure. Use M5 flat washers (51108390-006), M5 lock washers (51108884-006), and M5 thread-forming screws (51195168-616) to install the mounting brackets (51304256-100) on the cabinet rails as illustrated in Figure 4-7. The figure indicates where screws and washers must be initially installed during the first step in installing the brackets, and finally installed when mounting the UCNE enclosure on the brackets. Choose the illustration for either the right side or left side installation of the UCNE enclosure. The partially inserted screw can be used to initially support the enclosure while it is being mounted. Select the appropriate illustration and note the comments. The brackets are mounted reversed from each other. The upper bracket's tabs are oriented downward and the lower bracket's tabs are oriented upward.
2	Mount the UCNE enclosure to either the left or right side of the mounting brackets and secure it to the brackets with M5 flat washers, lock washers, and thread-forming screws. Refer to the appropriate illustration in Figure 4-7 for placement of the screws and washers. The partially inserted screw can be used to initially support the enclosure while it is being secured to the mounting bracket by other screws.
3	Attach the pair of UCNEs to mounting plates (51309266-100) with four M4 pan-head screws that include a captive external lock washer (51109695-308).
4	Place one of the UCNEs on the enclosure's upper shelf and secure it in place with the mounting plate's captive screw.
5	Place the second UCNE on the enclosure's third shelf from the top and secure it in place with the mounting plate's captive screw.
6	If two pair of UCNEs are being installed, repeat Steps 3 through 5, but place the third and fourth UCNEs on the second and fourth shelves from the top.
7	Connect a grounding wire (51201807-100) between the ground stud on the UCNE's front panel and the enclosure chassis. Use a M4 pan-head screw (51109695-308) to fasten it to the enclosure chassis.

4.9.3 UCNE Enclosure Installation Procedure, Continued

UCNE enclosure installation procedure, continued

Table 4-3 UCNE Enclosure Installation Procedure, Continued

Step	Action
8	Connect a power cable (51201397-002) between a power connector on the Power System's backpanel and the power connector at the front of each UCNE.
9	Installation of the UCNE enclosure(s) is complete. Use the procedures in Tables 4-4 and 4-5 to connect the UCNEs to the appropriate UCN coaxial segment and install the fiber optic cable(s) between the UCNEs.

UCNE testing and troubleshooting

Maintenance of the Universal Control Network Extender which includes testing and troubleshooting procedures for the UCNE can be found in the *Universal Control Network Guidelines* manual.

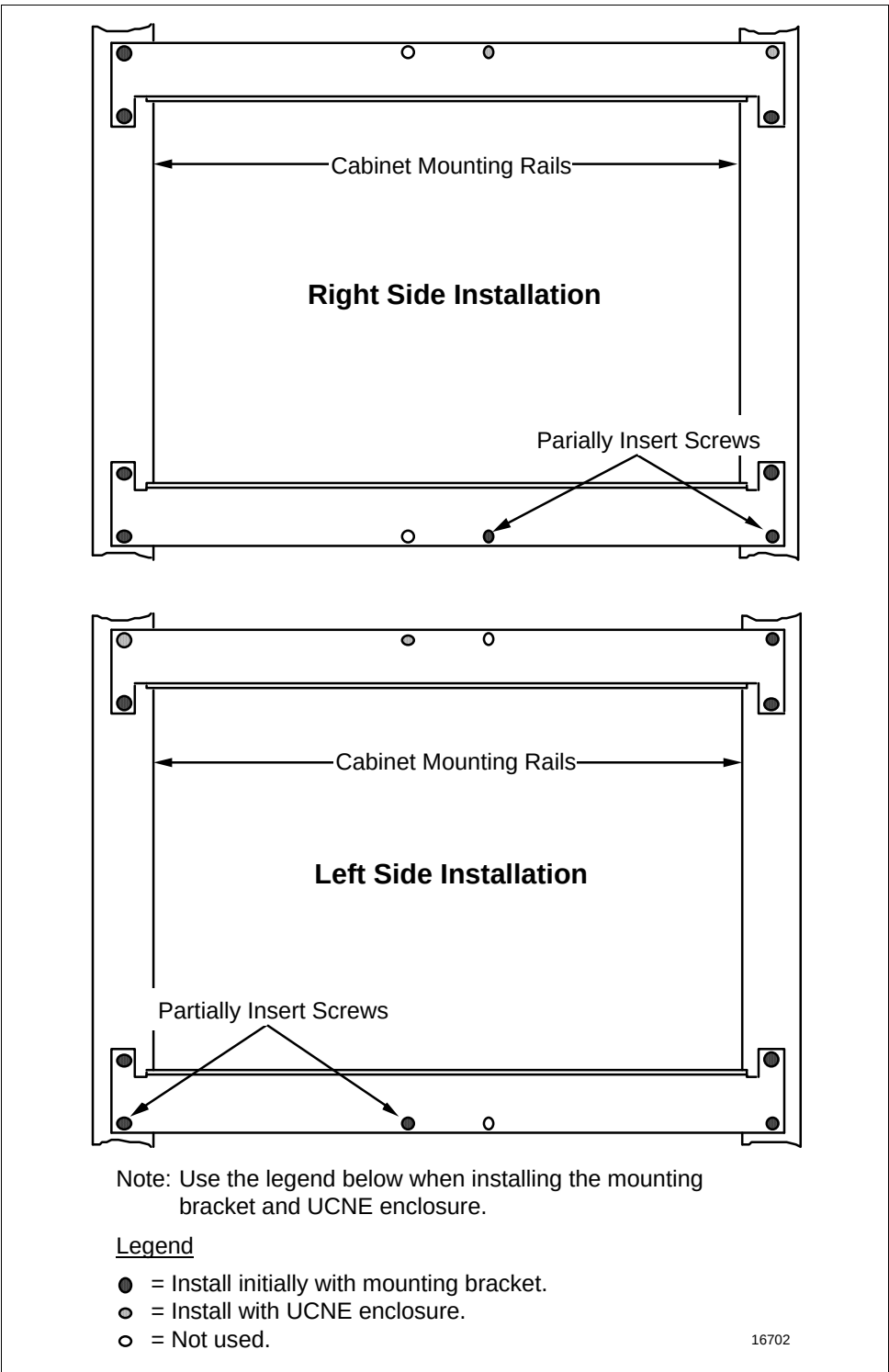
Because the amount of light that is emitted by the Light Emitting Diode (LED) into the fiber optic cable deteriorates over time, it is recommended that the UCNEs and the fiber optic link be tested periodically when convenient.

4.9.3 UCNE Enclosure Installation Procedure, Continued

UCNE enclosure
cabinet mounting
hardware

Figure 4-7 illustrates the use of the cabinet mounting hardware.

Figure 4-7 UCNE Mounting Bracket and Enclosure Installation



4.9.4 UCN Cable Installation Procedure

UCNE coaxial cable connection procedure

Use the procedure in Table 4-4 to connect the UCNEs to the UCN drop tap ports on the appropriate UCN coaxial trunk cable segments. The procedure assumes that new UCN coaxial trunk cable segments have been properly fabricated and tested, and the existing cable segments have been properly modified, if necessary, and tested by using information that is provided in Sections 2, 3 and 5 in this manual and the *Universal Control Network Planning* manual.

Table 4-4 UCNE Coaxial Cable Connection Procedure

Step	Action
1	Connect a UCN drop cable (RG6-type) between the UCN DROP (COAX) connector at the front of each UCNE and the appropriate UCN coaxial cable tap port. Observe the correct color code. Cable A has an orange-colored label and cable B has a brown-colored label. The UCNEs in positions 1 (UCN-A) and 3 (UCN-B) of the UCNE enclosure are a redundant pair, and optional UCNEs in positions 2 (UCN-A) and 4 (UCN-B) are another redundant pair.
2	Tighten the drop cable connectors to 25 inch/pounds by using the torque wrench that is included in the model MU-NKTQ01 kit.
3	Dress the drop cables (without violating the minimum bend radius rule) and secure them to cable anchors at either side of the enclosure.

4.9.5 Fiber Optic Cable Installation Procedure

UCNE fiber optic cable connection procedure

Use the procedure in Table 4-5 to install UCNE fiber optic cable. It is highly recommended that an experienced contractor fabricate, install, and test the fiber optic cables after installation.

The installed fiber optic link (optic fibers, connectors, splices, etc.) must meet the UCNE signal loss budget requirement. Reference information can be found in the *Universal Control Network Planning* manual.

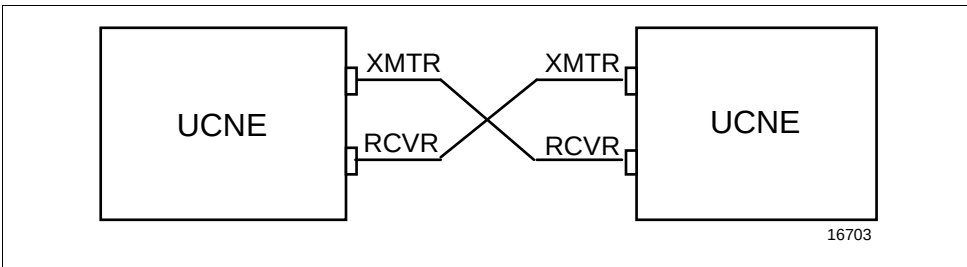
Table 4-5 UCNE Fiber Optic Cable Connection Procedure

Step	Action
1	Install the fiber optic cables between the pair of UCNEs at one end of the fiber optic link and the pair of UCNEs at the other end of the link. The UCNE enclosures can be resident in an LCN cabinet or console, in an LM cabinet, SM cabinet, or in a PM, APM, or HPM cabinet. The <i>Local Control Network Installation</i> manual details the installation of a UCNE enclosure in an LCN cabinet or console, an LM cabinet, or an SM cabinet.
2	Be sure the fiber optic cable connections are proper. The UCNE's Receive (RCVR) connector must be connected to the Transmit (XMTR) connector at the other end for both cable A and cable B pairs. Figure 4-8 illustrates the interconnections. Cables that are fabricated and labeled according to Honeywell drawing 51195498 can be readily and properly connected to the UCNEs.
3	Dress the cables and secure them to cable anchors at either side of the enclosure. It is extremely important not to compress the cable fibers or violate their minimum bend radius during or after clamping the cables. One method of properly securing the cables is to clamp the "strength member" that runs parallel to the optic fibers in many common fiber optic cables and not the optic fibers themselves. If a "strength member" is not present, it is also better to clamp the cable far enough from the end of the cable, before the individual optic fibers are "broken out."
4	Install the front panel (51201551-300) on the enclosure and secure it in place with its four fasteners.

Fiber optic cable interconnections

Figure 4-8 illustrates the UCNE fiber optic cable interconnections.

Figure 4-8 UCNE Fiber Optic Cable Interconnections



Section 5 – Network Checkout

5.1 Overview

Introduction

This section covers the checkout of the network after installation, or if an addition is made to the network at some later time.

Purpose

The following procedures provide instructions and references as needed for the initial checkout of the Universal Control Network (UCN) coaxial cables and drop cable taps. The instructions are intended to be understood by trained Honeywell or customer service technicians. The checkout procedures determine if the subsystem components have been properly installed and are ready for system startup. These procedures also serve to checkout system expansions or significant modifications. Please read this section completely before proceeding with the subsystem checkout.

Assembly instructions for the RG6-type drop coaxial cable and RG11-type UCN trunk coaxial cable are provided in *Section 3*. A list of required tools is also provided. Connector kits can be purchased from Honeywell. See the *Process Manager/Advanced Process Manager Service* or *High-Performance Process Manager Service* manual.

Appendix A

The form in Appendix A, the *PM/APM/HPM Post Installation Checklist*, should be used to monitor the progress and completeness of the checkout effort.

Appendix B

The form in Appendix B, *UCN Cable Measurements*, is to be used to record the characteristics of your UCN cable system when you check out your installation. The form can be copied or used as a guide in preparing your own form.

5.2 Tools and Test Equipment

Suggested tools and test equipment lists

The following is a suggested list of tools and test equipment that are needed to install and test a Universal Control Network.

Tools

- Standard tool kit
 - Test leads
 - Ground rods
-

Test equipment

- Dranetz Series 626 Universal Disturbance Analyzer*
 - Dranetz-BMI PowerVisa 440 Power Analyzer**
 - Series 626 Accessories (referenced for use in this manual):
 - 626-PA-6001-T AC Single-Phase Plug-in Monitor*
 - 626-PA-6003-T Three-phase AC Plug-in Monitor*
 - 626-PA-6002A-T Low Voltage DC Plug-in Monitor*
 - 626-PA-6020 RFI Measurement Plug-in Monitor*
 - Carrier Band Network Tester, Relcom CB Tester*
 - Accessories (provided with the Relcom CB Tester):
 - Relcom Auxiliary CB Signal Generator
 - Charger/Power Plug
 - Probe Cable (or MU-MCA002)
 - 1% 75 Ohm Terminator, etc.
 - Biddle Megger® Earth Tester, Model 250200, 250220 or 250241*
 - Fluke1625 Kit or AEMC 3731 Clamp-on Ground Resistance Tester ***
- * TPS system test devices should be available as needed. You can choose equipment equivalent to that specified, but test procedures and results may vary.
- ** New test devices. Dranetz-BMI PowerVisa 440 can be used instead of Dranetz Series 626 Analyzer, and it does not need dc plug-in modules.
- *** New test devices. Fluke1625 Kit or AEMC 3731 can be used instead of the Biddle Megger Earth Tester.
-

ATTENTION

Two CB testers are required if an Auxiliary CB Signal Generator is not provided.

ATTENTION

Biddle Model 250260 is not acceptable.

5.3 UCN AC Safety Ground

**Drop cable tap
grounding**

Visually inspect the UCN cabling to verify that every drop cable tap on each UCN trunk cable is connected to the nearest AC Safety Ground with 2.5 mm² (14 AWG) wire as detailed in subsection 2.5.

Ground measurement

Using the Earth Tester, measure less than 0.5 ohm between the UCN drop cable taps and the facility's AC Safety Ground.

5.4 UCN Testing

Introduction	Refer to Figure 5-1 for the following discussion. The Universal Control Network (UCN) uses carrier-band hardware for its implementation. A redundant 2-trunk system, as illustrated, must always be installed.
Typical UCN	A typical UCN cable system may consist of two trunk cables with four drop cable taps on each trunk as shown in Figure 5-1. Each tap can provide network connections for two (or four, or eight) Universal Control Network devices. The UCN device can be a primary or redundant Process Manager Module (PMM), a primary or redundant Advanced Process Manager Module (APMM), a Logic Manager (LM), or a Network Interface Module (NIM). Redundant network devices must share the same drop cable taps.
External electrical noise	Sufficient signal strength throughout the network is necessary for the UCN devices to communicate properly. The network should also be checked for excessive external electrical noise that may interfere with communications.
Network parameters	All network parameters are measured in decibel millivolts (dBmV) of signal strength or decibels (dB) of attenuation. These parameters are easy to measure because they are additive, or subtractive in nature. Use a Relcom Carrier-Band Network Tester to verify the quality of the Universal Control Network. Table 5-1 lists the strengths and losses for UCN components.

5.4 UCN Testing, Continued



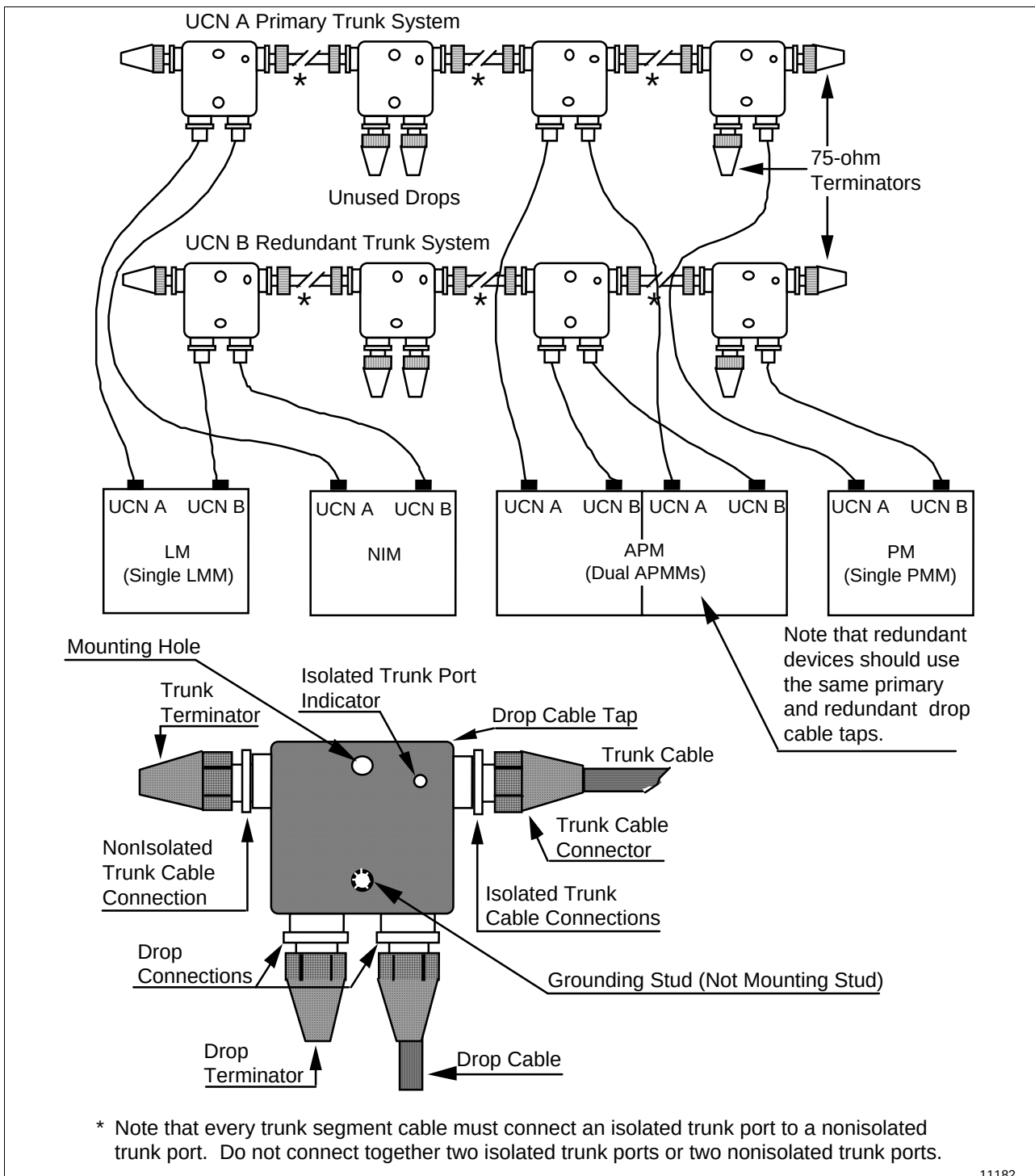
SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

5.4 UCN Testing, Continued

Typical UCN cable system

Figure 5-1 Typical UCN Cable System



11182

5.4 UCN Testing, Continued

UCN component parameters

Table 5-1 is a parameter list of UCN components.

Table 5-1 UCN Component Signal Strengths and Losses

Component	Strength/Loss	Parameter
UCN Transmitter	64.5±1.5 dBmV	Nominal output signal strength
UCN Receiver	10.0 dBmV	Minimum signal strength sensitivity
UCN Drop Cable Tap	0.5 dB	Maximum trunk pass through attenuation
	20.0 dB	Nominal drop attenuation loss
UCN RG11-Type Trunk Cable	1.5 dB	Attenuation per 100 meters (328 ft)*
UCN RG6-Type Drop Cable	2.5 dB	Attenuation per 100 meters (328 ft)*

* Values for standard UCN cable at 5 Mb/s. This can vary depending upon the cable type.

Tester accuracy

When using the carrier-band tester, keep in mind the accuracy of the tester when comparing actual measurements with Table 5-1.

Component attenuation

Although it is necessary for isolation, each drop cable tap produces a 20 dBmV signal loss in each direction between the drop cable port and the trunk cable port. There is very little signal attenuation between the isolated trunk port and the nonisolated trunk port in the tap. Attenuation varies through the cables, but must not be more than an additional 13 dB for any transmitter to receiver path, including trunk cable, drop cable, and pass-through losses.

Record measurements

Keep a record of your measurements. You can copy the sample in Appendix B of this manual, or use it as a guide in preparing your own form.

WARNING

When disconnecting and reconnecting cables in the procedures that follow, be aware of a possible shock hazard because of shield potential differences between cables. Cables can become ungrounded and present a shock hazard because of the possible ground potential differences in various plant areas.

5.4 UCN Testing, Continued

CAUTION

CAUTION—The primary trunk cable system must be connected to all cable A connectors and the redundant trunk cable system connected to all cable B connectors. This convention must be maintained throughout the Universal Control Network.

The ends of the trunk cables and all unused drop cable connections on the taps must be terminated with carrier-band rated 75-ohm terminations. Visually check the system to verify that all terminations and cable connections are in place and properly tightened.

When initially tightening terminations and cable connectors, turn the connectors finger tight, then wrench-tighten the connectors to 25 inch/pounds with a torque wrench. Do not overtighten the connectors. A slight twist on the cable will indicate if the connection is tight. If the connection is not proper, retighten the connector with the torque wrench.

Avoid stress on cables where they connect to the drop cable taps or network device UCN connectors. Do not compromise cable shield integrity by violating the minimum bend radius. If the connectors are not properly installed, the Universal Control Network may be degraded and poor system performance can result.

Network test procedures

The test procedures that are described in the following subsections should be performed after the network devices have been wired and cabled to the network according to the procedures in their respective installation manuals. The power circuit breakers for the devices must be in the off position.

5.4.1 Ambient Noise Testing

Ambient noise testing procedure

Ambient electrical noise is present in any electrical environment. Although the UCN cable system guards against noise, check the installed network to verify that noise does not interfere with Universal Control Network communications.

Use the procedure in Table 5-2 to determine the ambient noise level.

Table 5-2 Ambient Noise Testing Procedure

Step	Action
1	Be sure all terminators and all network devices are connected to the cable system and power is removed from all the network devices. Verify that the 48 V backup battery system, if present in the Process Manager or Advanced Process Manager, is not maintaining power. Check that the 48 V Battery switch is in the off position.
2	Remove a drop cable or 75-ohm terminator from any drop cable port on the tap at one end of the primary trunk cable of the Universal Control Network under test and connect the Relcom CB Tester to the port. Set the tester to NOISE and 5 Mb/s . Move the PEAK/RESET switch to the RESET position momentarily and then leave it in the PEAK position. Do not use the ac adapter provided with the Relcom tester while making noise measurements. The tester must operate with its internally charged batteries. Read the tester after 40 or more seconds. Verify and record a noise figure less than -10 dBmV (example: -11 dBmV is less than -10 dBmV, -9 dBmV is not). If the measurement is less than -25 dBmV, the tester will flash "-25." This is a satisfactory reading.
3	Disconnect the CB tester, reconnect the drop cable or 75-ohm terminator, and move the tester to the redundant trunk cable of the Universal Control Network under test. Repeat Step 2 for the redundant trunk, then disconnect the tester and reconnect the cable or terminator.
4	Go to the other end of the trunk cables of the same Universal Control Network and repeat Steps 2 and 3.

Analyzing a test failure

If your system fails to pass the ambient noise test in this subsection, thoroughly check the cabling and drop cable tap grounding systems. Also, attempt to find the source of any electrical noise interference. Data obtained from any RFI tests conducted will help. If no interference was found in those tests, try setting the Dranetz test equipment to a lower sensitivity or moving the test equipment to locations nearer the Universal Control Network trunk and drop cables. In some cases, consider substituting armored trunk cable in place of standard trunk cable.

5.4.2 UCN Cross Connection Test

Cross connection test procedure Perform an electrical check to verify that the UCN A and B trunk cables are not cross-connected by using the procedure in Table 5-3.

Table 5-3 Trunk Cable Cross Connection Test Procedure

Step	Action
1	Be sure all 75-ohm terminators and all UCN devices are connected to the cable system, and all UCN devices have power removed. Verify that the 48 V backup battery system, if present in the Process Manager or Advanced Process Manager, is not maintaining power. Check that the 48 V Battery switch is in the off position.
2	Disconnect the A drop cable from a UCN device at one end of the primary trunk cable and connect the cable instead to the Relcom Auxiliary CB Signal Generator, or another CB Tester set to the SIG GEN position, using the XMIT port. The signal generator will transmit a signal along the trunk cable system that will be received by a tester that is connected to points along the primary UCN trunk cable system.
3	Remove the B drop cable from the UCN connection at the same end of the redundant trunk cable serving the same network device as in step 2. Connect the Relcom CB Tester to the drop cable. Set the tester switches to the NOISE and 5 Mb/s positions. Move the PEAK/RESET switch to the RESET position momentarily and then leave it in the PEAK position. Read the tester after 40 or more seconds. As in step 2 of subsection 5.3.1, do not use the ac adapter provided with the Relcom tester while making noise measurements. Check that the noise figure is less than -10 dBmV (example: -11 dBmV is less than -10 dBmV, -9 dBmV is not). If the noise test in Step 3 fails, there is probably an error in the cabling of the UCN cable system. A UCN B device is probably cross-connected to a UCN A device. Examine all the drop cable taps and network device connections until you find and correct the problem. Do not proceed to the next test until the problem is corrected.
4	Reconnect the drop cable to the UCN B device. Leave the auxiliary signal generator connected to the UCN A drop cable in preparation for the tests that follow.

5.4.3 UCN Return Loss and Signal Strength Tests

Introduction

Use the CB tester to measure return loss (in one direction) and the signal strength. In the steps that follow, it is unnecessary to measure return loss at each drop cable tap port to a UCN device. Measure the return loss only once at each tap. Use the UCN Cable Measurements form in Appendix B. If the return loss measurements were recorded during cable installation, a quick verification is sufficient.

Return loss and signal strength test procedure

You should measure and record signal strength at each UCN device. The "Expected Signal Strength" on the UCN Cable Measurements form in Appendix B can be calculated from attenuation measurements taken during installation. Simple addition and subtraction of dB readings is all that is necessary.

Use the procedure in Table 5-4 to measure and record the UCN cable signal return loss and signal strength at each device on the network.

Table 5-4 UCN Cable Return Loss and Signal Strength Test Procedure

Step	Action
1	With the Auxiliary CB Signal Generator still connected to the UCN A drop cable noted in step 4 of subsection 5.4.2, go to the other end of the primary trunk cable, remove the drop cable from a UCN A device at that end and connect the Relcom CB Tester to the removed drop cable.
2	Set the tester to RET LOSS, 5 Mb/s , and RESET . Verify and record a return loss of 14 dB or greater.
3	Set the tester to SIG LEVEL, 5 Mb/s , and RESET . Verify and record a signal level of +10 dBmV or greater.
4	Disconnect the drop cable from the tester and reconnect it to the UCN A device.
5	Move the tester to the next UCN A device on the primary trunk cable that is closer to the signal generator. Disconnect the drop cable, connect the tester to the drop cable, and repeat Steps 2 through 4.
6	Repeat Step 5 until all the UCN A device ports on the primary trunk have been checked for return loss and signal level.
7	Move the signal generator from the UCN A drop port to the corresponding UCN B drop port on the redundant trunk cable. Reconnect the UCN A device to its drop cable.
8	Repeat Steps 1 through 6, substituting UCN B devices and the redundant trunk cable for UCN A devices and the primary trunk cable. Reconnect all network devices to their drop cables when you have finished testing.

5.5 UCN Power-On Test

Introduction

This test procedure verifies that all transmitters in the Universal Control Network are quiet when power is applied to the equipment and the transmitters are idle or not transmitting.

Power-on test procedure

The power-on test can be made only when the Network Interface Module (NIM) and all of the process control equipment connected to the primary and redundant UCN trunk cables have power applied, but before software has been downloaded to the Network Interface Module through the LCN.

Use the procedure in Table 5-5 to perform a power-on test.

Table 5-5 UCN Power-On Test Procedure

Step	Action
1	Verify that no UCN device connected to the network is transmitting. Check the NIM and observe the TRANSMIT indicator on the PNI/EPNI board. Check the Modem card(s) of any PM/APM to verify that the POWER indicator is on but the STATUS indicator is off. Check the Comm/Control card(s) of any HPM to verify that the POWER indicator is on but the STATUS indicator is off. Check the Logic Manager Module card(s) of any LM to verify that the PASS indicator is on but the XMIT indicator is off. Observe each indicator continuously for approximately 20 seconds to ensure it doesn't flash. A flash indicates it is transmitting. If any of these indicators continues to flash, check the NORMAL-TEST jumper on the PNI/EPNI board of the NIM. Ensure that it is in the NORMAL position. Also, check the LFAC-LNET-RFAC-RNET jumper locations on the PMM or APMM backpanel of any PM or APM. No jumpers should be in these locations. Check again that there is no downloading of software to the NIM or that none has been downloaded. If the NIM was downloaded, remove power and then reapply power to the equipment. The downloaded program will be destroyed.

5.5 UCN Power-On Test, Continued

Power-on test procedure, continued

Table 5-5 UCN Power-On Test Procedure, Continued

Step	Action
2	Go to the drop cable tap at one end of the primary trunk cable of the Universal Control Network under test. Remove a drop cable or a 75-ohm terminator from any port connection on the tap, and replace it with the Relcom CB Tester. Set the tester switches to NOISE and 5 Mb/s positions. Move the RESET/PEAK switch to RESET position momentarily and then leave it in the PEAK position. Do not use the ac adapter provided with the Relcom tester while making noise measurements. See subsection 5.3.1.1, Step 2. Read the tester after 40 or more seconds. Verify and record a noise figure less than -10 dBmV (Example: -15 dBmV is less than -10 dBmV, -5 dBmV is not). If the measurement is less than -25 dBmV, the tester will flash "-25." This is a satisfactory reading.
3	Disconnect the CB tester, reconnect the drop cable or 75-ohm terminator, and move the tester to the redundant trunk cable of the Universal Control Network under test. Repeat Step 2 for the redundant trunk cable, and then disconnect the tester and reconnect the drop cable or 75-ohm terminator.
4	Go to the other end of both trunk cables of the same Universal Control Network and repeat Steps 2 and 3.

5.6 EUCN Power-On Test

Introduction

This test procedure verifies that all the transmitters in the EUCN are active when power is applied to the equipment.

Power-on test procedure

The power-on test can be made only when the Enhanced Network Interface Module (ENIM) and all of the process control equipment connected to the primary and redundant EUCN cables have power applied, but **prior to the** software download to the ENIM through the LCN.

Use the procedure in Table 5-6 to perform the EUCN power-on test.

Table 5-6 EUCN Power-On Test Procedure

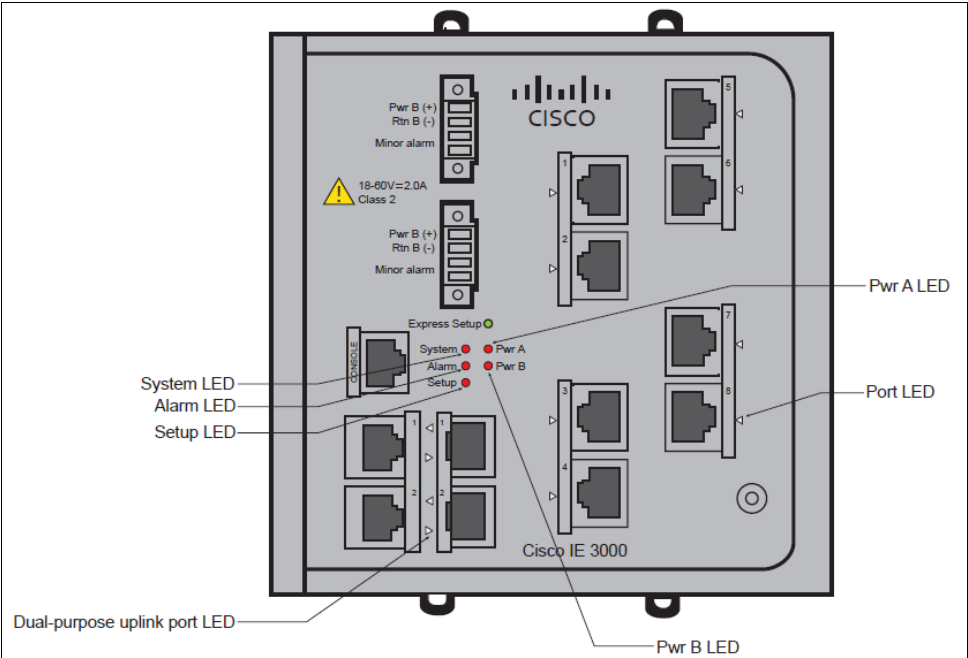
Step	Action
1	Power up all the Ethernet switches, CF9s, ENIMs and EHPMs that are a part of the test.
2	Check the CF9 and the Ethernet switches and observe the Power indicator is lit. Note: You may observe other LED indicators that are flashing. Check the ENIM and observe that the LINK indicator on both the channels of the FTE cable A and B are lit on the EPNI2 board.
3	Check the ENIM and observe that the TRANSMIT indicator on both the channels of the FTE cable A and B are blinking on the EPNI2 board.
4	Check the EHPM FTE Interface Module and observe that the LINK indicator on both the channels of the FTE cable A and B are lit.
5	Check the EHPM FTE Interface Module and observe the TRANSMIT indicator on both the channels of the FTE cable A and B are blinking.
6	Check all the ENIM and EHPM cards for the parameters mentioned above.

5.7 FTE Switch Status indicators

Interpretation of Cisco IE 3000 Front Panel & Port LED Indicators

You can use the LEDs to monitor the switch status, activity, and performance. The figure below shows the front panel LEDs, and the following sections describe them.

Figure 5-2 FTE Switch LEDs



Initial Switch set up using “Express Setup”

Table 5-7 Initial Switch set up using Express Setup – System LED

LED Color	Description
Blinking green	The switch power-on self-test (POST) is running.
Solid green	POST has completed.
Red	The switch has failed POST.
Blinking green	The switch power-on self-test (POST) is running.

Table 5-8 Initial Switch set up using Express Setup – Setup LED

LED Color	Description
Blinking green	The switch is not configured.
Solid green	Start a browser session on the PC.

Table 5-9 Initial Switch set up using Express Setup – Port LED

LED Color	Description
Blinking green	The switch Express Setup button is pressed. A Category 5 Ethernet cable connects the blinking switch port to the Ethernet port on a PC.

Table 5-10 Initial Switch set up using Express Setup – Pwr A LED

LED Color	Description
Solid green	The switch is properly powered up.
Dark (Off)	The switch is not powered up.

Verifying Port Connectivity

After the switch port is connected to another device.

Table 5-11 Verifying Port Connectivity – Port LED

LED Color	Description
Amber	The switch is establishing a link to the connected device. This process takes about 30 seconds.
Solid green	The switch and target device have established a link.
Dark (Off)	The target device might not be turned on or there might be a cable problem.

Normal Operating Mode

The switch is completely configured and connectivity with the target device is already established.

Table 5-12 Normal Operating Mode – Port LED

LED Color	Description
Flashing green	Communication between the port and target device is occurring
Flashing green interspersed with flashing amber OR solid amber	A problem exists with this port to target device interface. Check the port configuration and cable integrity. If OK and condition still persists, contact Honeywell Technical Assistance Center.
Flashing amber	Communication with the target device has failed.
Dark (Off)	The target device might not be turned on or there might be a cable problem.

Table 5-13 Normal Operating Mode – System LED

LED Color	Description
Green	Indicates normal operation.
Red or amber	There might be a hardware problem. Contact the Honeywell Technical Assistance Center.

Table 5-14 Normal Operating Mode – Setup LED

LED Color	Description
Dark (Off)	Indicates normal operation.
Red or amber	There might be a hardware problem. Contact the Honeywell Technical Assistance Center.

Table 5-15 Normal Operating Mode – Alarm LED

LED Color	Description
Green	Indicates normal operation.
Red or amber	A preconfigured alarm set point has been exceeded. Contact the Honeywell Technical Assistance Center.

Table 5-16 Normal Operating Mode – Pwr A LED

LED Color	Description
Solid green	The switch is properly powered up.
Red or amber	The switch is not powered up. Contact the Honeywell Technical Assistance Center.

Table 5-17 Normal Operating Mode – Pwr B LED (Applies only if redundant power sources are connected.)

LED Color	Description
Solid green	The switch is properly powered up.
Red or amber	The switch is not powered up. Contact the Honeywell Technical Assistance Center.

5.8 Control Firewall Status indicators

Control Firewall Startup

The following table summarizes the stages the CF9 goes through after power is applied to its IOTA during startup. The CF9 repeats these stages every time power is cycled Off/On or the pins on its reset pad are shorted.

Table 5-18 Control Firewall Status indicators

Stage	Description
1	Power Light Emitting Diode (LED) lights (green).
2	Status LED is red while the CF9 runs its power-on self test (POST).
3	When the CF9 POST completes, the Status LED turns green and the FTE port LEDs blink green and off for all connected ports when there is traffic, and remain steady green for connected ports when there is no traffic. The LEDs for unconnected FTE ports are off.

CF9 LED Descriptions

The following illustration and table identify and describe the indicators associated with the LEDs on the CF9.

Figure 5-3 CF9 LEDs



Table 5-19 CF9 LED Descriptions

LED	State				
	Off	Green	Red	Blinking green	Green with brief flicker
Power	No Power	Power On	N/A	N/A	N/A
Status	No Power of Fault	Normal Operation	Running POST	*See Note	N/A
Downlink Ports 1 to 8	No Power or No Cable Connected	Link Present, No Traffic	N/A	Link Present, Traffic	Too Many Attached Devices
Uplink Port	No Power or No Cable Connected	Link Present, No Traffic	N/A	Link Present, Traffic	N/A
Note: The CF9 has detected a soft failure and is working in a diminished state. Typical causes are a disconnected uplink port cable or a downlink port with too many attached devices. (Only one device can be attached to a CF9 downlink port.)					

Appendix A – PM/APM/HPM/EHPM Post-Installation Checklist

Customer _____ Subsystem ID _____ Date (Delivery) _____ Date (Start-up) _____

Test	Description	Done?
PRE-POWER CONNECTION TESTS	Use procedures in the appropriate Installation and Checkout manuals.	
General Site-Preparation Checks	Check the facility's AC Safety Ground earth resistance with an Earth Tester. Check that the AC power is the proper voltage. Check the AC line stability with Disturbance Analyzer. Do an RFI measurement where any UCN/EUCN device is installed.	
POWER-OFF CHECKS	Use procedures in the appropriate Installation and Checkout manuals.	
Grounding Checks	Inspect visually and physically for completeness. Check that local AC Safety Ground is less than 0.5 ohm (also check the cabinets without ac power). Check the local Master Reference Ground (MRG). Do an MRG noise test if a substitute MRG is used. Check the local Zener Barrier Ground bus connection. Check the UCN Safety Ground connections at all taps.	
UCN/EUCN Cable Visual Checks	For UCN, Ensure that primary cables connect to A connectors and redundant cables to B connectors at all taps. (Note the cables are marked with an identifying color code. An Orange Label designates the A cable and a Brown label designates the B Cable) Verify that the terminators are installed on all unused tap connections. For EUCN Ensure that primary FTE cables connect to A connectors and redundant cables to B connector on all Switches ENIMs and EHPMs (Note the cables are marked with an identifying color code. A yellow boot designates the A cable and a Green boot designates the B Cable)	
UCN/EUCN Power-Off Cable Tests	For UCN, see Section 5 in this manual and check the completed UCN recorded noise, return loss, and signal strength. For EUCN, visually check that the cable connectors are fully engaged in the appropriate NIM, Switch, CF9 or EHPM interface connector. Verify that all of the EUCN A cables (yellow cables) are routed to A cable connections on CF9s or Level1/Level2 switches. Repeat the check for the B cable (green cables). Identify Control Firewalls and switches associated with the A cable and verify that only A cables (Yellow Boots) connect to these devices. Identify Control Fire Walls and switches associated with the B cable and verify that only B cables (Green Boots) connect to these devices. Note dependent upon which device is the highest level switch in the network, there may be a White crossover cable attached to this switch.	
General Pre-Power Checks	If applicable, check the AC Voltage Source Selection switch on the Power Supply Module(s) in the PM, APM, or HPM.	
POWER-ON CHECKS	Check the procedure in the appropriate Installation or Checkout manual.	

AC Power Verification	Verify operation of Power Supply Module fans. If the system is 120 volt ac or 240 volts ac with neutral, check that the voltage is between 100 Vac and 132 Vac. Check for a measurement of no more than a 0.5 volts between neutral and the cabinet frame. If the system is 240 volt ac, check that the voltage is between 187 Vac and 264 Vac.	
Backup Power Verification	Verify the Power System 48 V Battery Backup system (optional) is operational, and the CMOS battery backup voltage is proper for any PM, APM, or HPM.	
Test	Description	Done?
Power Supply Output Verification	Check that all green LEDs on Power Supply Modules are lit. Check that the 24 volt dc output is between 23 Vdc and 25 Vdc. Check that the 6 volt ac rms output is between 2 Vac and 7.2 Vac.	
Card Self-Test Verification	Check each card module for proper green Power and Status indicator status.	
UCN/EUCN Power-On Cable Tests	For UCN, check and record the noise figure of the UCN with all nodes powered and connected, but silent. Verify the UCN address pinning (See the appropriate Installation manual). For EUCN, see section 5.6 EUCN Power-On Test	
Network Test Mode Checks (UCN)	Check that all UCN devices engage in token-passing operation.	

Appendix B – UCN Cable Measurements

Customer _____ Subsystem ID _____ Date (Delivery) _____ Date (Start-up) _____
 Network Name _____ Number _____ Cable Segment _____

Tap Number	Building Location	Drop Name or Usage	Noise Figure	Return Loss		Signal Strength	
				"A" Dir	"B" Dir	Expected	Measured
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____
_____	_____	1. _____ 2. _____	_____	_____	_____	_____	_____

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